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Environmental Pollution Detection using Deep Learning

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Αριστοτελειο Πανεπιστημιο Θεσσαλονικης

Ανίχνευση περιβαλλοντικής μόλυνσης με χρήση βαθιάς μάθησης

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Περίληψη

Οι ραγδαίες εξελίξεις των τελευταίων ετών στον κλάδο της τεχνητής νοημοσύνης έχουν οδηγήσει σε μεγάλες αλλαγές σε όλους τους τομείς της ζωής μας. Η Βαθιά Μάθηση (Deep Learning), δηλαδή ο τομέας της τεχνητής νοημοσύνης που χρησιμοποιεί νευρωνικά δίκτυα, θεωρείται πλέον από τις πιο αποτελεσματικές προσεγγίσεις για την επίλυση σύνθετων και εξαιρετικά δύσκολων προβλημάτων, των οποίων η προσπάθεια επίλυσης με απλά μαθηματικά και στατιστικά μοντέλα φαίνεται να αποτυγχάνει. Η επιτυχία της βαθιάς μάθησης οφείλεται κυρίως στην ικανότητα της να εκμεταλλεύεται το τεράστιο πλήθος δεδομένων που υπάρχουν διαθέσιμα στην σημερινή εποχή και να μαθαίνει από αυτά, με αποτέλεσμα να αυξάνεται η ακρίβεια των προβλέψεων της.

Η βαθιά μάθηση χρησιμοποιείται ευρέως για την ανίχνευση περιβαλλοντικής ρύπανσης, καθώς επιτρέπει την ανάπτυξη μοντέλων που μπορούν να αναγνωρίζουν και να ταξινομούν αέρια ρύπανσης από διάφορες πηγές δεδομένων. Οι αλγόριθμοι βαθιάς μάθησης προσπαθούν να βρουν τις βέλτιστες παραμέτρους που μεγιστοποιούν την ακρίβεια ανίχνευσης, μαθαίνοντας πολύπλοκα μοτίβα από τα δεδομένα εκπαίδευσης.

Μια ιδιαίτερα ελπιδοφόρα τεχνολογία για την ανίχνευση περιβαλλοντικής ρύπανσης είναι τα φωτονικά φασματόμετρα Spatial Heterodyne Fourier-Transform (SHFT), τα οποία προσφέρουν συμπαγείς και οικονομικές λύσεις ανίχνευσης αερίων. Η χρήση της βαθιάς μάθησης σε συνδυασμό με τα φασματόμετρα SHFT μπορεί να οδηγήσει σε ακριβή μοντέλα ταξινόμησης αερίων. Ωστόσο, υπάρχουν σημαντικές προκλήσεις που πρέπει να αντιμετωπιστούν: τα φασματοσκοπικά δεδομένα από φωτονικές συσκευές αποτελούν ένα ιδιαίτερα θορυβώδες, μεταβαλλόμενο και εξαρτώμενο από το υλικό περιβάλλον. Αυτό οφείλεται στην φύση των φασματοσκοπικών μετρήσεων, στις θερμοκρασιακές διακυμάνσεις που επηρεάζουν τις συσκευές, καθώς και στα σφάλματα κατασκευής που εισάγουν συστηματικές παραμορφώσεις και διαφορές μεταξύ των συσκευών ανίχνευσης.

Οι προκλήσεις αυτές καθιστούν δύσκολη την εκπαίδευση σταθερών μοντέλων που μπορούν να εκτελούν αξιόπιστες μετρήσεις σε πραγματικές συνθήκες. Συγκεκριμένα, μοντέλα που εκπαιδεύονται σε μια συγκεκριμένη φωτονική συσκευή παρουσιάζουν συχνά σημαντική υποβάθμιση της απόδοσης τους όταν εφαρμόζονται σε διαφορετικά τσιπ, λόγω των διαφορών στα φασματοσκοπικά χαρακτηριστικά που προκαλούνται από τα σφάλματα κατασκευής. Παράλληλα, οι θερμοκρασιακές διακυμάνσεις εισάγουν επιπλέον παραμορφώσεις που επηρεάζουν την αξιοπιστία της ταξινόμησης. Επομένως, κρίνεται απαραίτητη η ανάπτυξη μεθοδολογιών

που επιτυγχάνουν διασκευασιακή γενίκευση (cross-hardware generalization), δηλαδή έχουν την δυνατότητα να προσαρμόζουν την συμπεριφορά τους με βάση τις διαφορές μεταξύ διαφορετικών φωτονικών συσκευών και θερμοκρασιακών συνθηκών. Το βασικό πρόβλημα που ανακύπτει είναι η ισορροπία μεταξύ αποθορυβοποίησης και διατήρησης πληροφορίας: το μοντέλο πρέπει να μειώνει τις παραμορφώσεις από θερμοκρασία και σφάλματα κατασκευής, διατηρώντας παράλληλα τις κρίσιμες πληροφορίες που χρειάζονται για την ακριβή ταξινόμηση των αερίων.

Στην διπλωματική αυτή προτείνεται μια μεθοδολογία διαδοχικών αυτοκωδικοποιητών (sequential autoencoders) για την ανίχνευση περιβαλλοντικής ρύπανσης με διασκευασιακή γενίκευση. Συγκεκριμένα, εκπαιδεύονται δύο διαδοχικοί αυτοκωδικοποιητές για κάθε τσιπ: ένας για αντιστάθμιση θερμοκρασίας και ένας για αντιστάθμιση σφαλμάτων κατασκευής. Οι αυτοκωδικοποιητές εκπαιδεύονται αρχικά ανεξάρτητα, διατηρώντας ταυτόχρονα τόσο την ικανότητα αποθορυβοποίησης όσο και τις διακριτικές πληροφορίες για την ταξινόμηση. Στην συνέχεια, εκπαιδεύεται ένας καθολικός ταξινομητής που λειτουργεί σε όλα τα τσιπ ταυτόχρονα, ενώ όλοι οι αυτοκωδικοποιητές ρυθμίζονται λεπτομερώς (fine-tuning) μαζί με τον ταξινομητή. Με τον τρόπο αυτό, οι αυτοκωδικοποιητές προσαρμόζουν τις αναπαραστάσεις τους με βάση την τελική απόδοση ταξινόμησης, επιτυγχάνοντας βέλτιστη ισορροπία μεταξύ αντιστάθμισης παραμορφώσεων και διατήρησης πληροφορίας.

Επιπρόσθετα, χρησιμοποιήθηκε κανονικοποίηση βασισμένη στην κλάση αναφοράς (reference-based normalization) με robust scaling που κανονικοποιεί κάθε μέτρηση στα στατιστικά της κλάσης αναφοράς. Η τεχνική αυτή χρησιμοποιεί τη διάμεσο (median) και την Median Absolute Deviation (MAD) από την κλάση αναφοράς, μειώνοντας την επιρροή βασικών διακυμάνσεων και τονίζοντας τα διαφορικά χαρακτηριστικά που εισάγει το αέριο.

Η προτεινόμενη μεθοδολογία, αποδείχτηκε να υπερτερεί συστηματικά στην ανίχνευση περιβαλλοντικής ρύπανσης σε σύνολο δεδομένων πολλαπλών τσιπ με διακυμάνσεις θερμοκρασίας και σφάλματα κατασκευής. Φαίνεται, επομένως, η συνδυασμένη προσέγγιση διαδοχικών αυτοκωδικοποιητών με κοινή λεπτομερή ρύθμιση να βελτιώνει σημαντικά την απόδοση μοντέλων που αλληλεπιδρούν σε διαφορετικά περιβάλλοντα υλικού, και να επιτρέπει στα μοντέλα να διατηρούν υψηλή απόδοση παρά τις διαφορές μεταξύ των φωτονικών συσκευών ανίχνευσης και τις διακυμάνσεις θερμοκρασίας.

Abstract

The rapid increase of environmental pollution has accentuated the need for advanced monitoring systems that can accurately detect and classify pollutants across diverse sensor platforms and environmental conditions. Deep learning has been proven well-suited for this task, as it inherently aligns with the goal of environmental monitoring i.e. maximizing detection accuracy and reliability across varying hardware and temperature conditions. Among emerging sensing technologies, Spatial Heterodyne Fourier-Transform (SHFT) spectrometers offer promising capabilities for compact, low-cost gas detection through on-chip interferometric measurements. However, measurements from photonic sensing devices often exhibit noisy, variable, and hardware-dependent characteristics that pose challenges for standard deep learning methods. Therefore, the implementation of sophisticated compensation strategies is necessary for models to successfully generalize across different sensing hardware configurations. In this work, we propose a novel sequential autoencoder methodology that incorporates explicit temperature and fabrication error compensation mechanisms to balance the trade-off between noise reduction and preservation of critical spectral signatures, enabling the model to accurately detect gas absorption across diverse chip platforms and temperature ranges. The efficacy of our proposed method is amplified by a multi-task learning strategy used during independent autoencoder training, which helps preserve class-discriminative features while removing hardware-specific distortions. Additionally, we leverage joint optimization techniques across all model components by employing simultaneous fine-tuning of chip-specific autoencoders and a global classifier. This approach enables the model to be calibrated for optimal cross-chip generalization while adapting representations to maximize classification performance. In our comprehensive evaluation using interferometric data from multiple simulated SHFT spectrometer chips with fabrication errors and temperature variations, we demonstrate consistent improvement in classification accuracy across all chips and conditions, with special emphasis on maintaining excellent generalization and robust baseline detection capabilities for clean environmental conditions.

Ευχαριστίες

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1

Introduction

Environmental monitoring has become increasingly important as concerns about air quality and pollution continue to grow. The accurate detection and classification of environmental pollutants requires advanced sensing technologies combined with sophisticated computational methods. Among emerging sensing technologies, Spatial Heterodyne Fourier-Transform (SHFT) spectrometers offer promising capabilities for compact, low-cost gas detection through on-chip interferometric measurements using Mach-Zehnder Interferometer (MZI) arrays.

Despite their potential, photonic spectrometers face significant challenges that limit their practical deployment. Fabrication imperfections introduce systematic variations across different chips, causing each device to exhibit unique spectral response characteristics. Additionally, temperature fluctuations induce thermo-optic effects that distort the measured interferometric signals. These chip-to-chip variations and temperature-dependent distortions severely degrade classification performance, particularly when attempting to deploy models trained on one chip to operate on different chips under varying environmental conditions.

Traditional machine learning approaches struggle to generalize across chips with different fabrication characteristics and operating temperatures. While conventional normalization techniques can partially address intensity variations, they fail to correct for the systematic wavelength shifts and phase distortions caused by fabrication errors and temperature changes. This fundamental limitation has hindered the widespread adoption of photonic spectrometers for real-world environmental monitoring applications.

To address these challenges, this thesis proposes a deep learning framework based on disentangled autoencoders that explicitly compensate for both temperature-induced and fabrication-induced variations in SHFT spectrometer data. The proposed method consists of four key stages: (i) reference-based normalization that anchors measurements

to no-gas reference statistics, (ii) a temperature compensation autoencoder that maps measurements acquired at arbitrary temperatures to a fixed reference temperature, (iii) a fabrication error compensation autoencoder that aligns chip-specific responses to a common ideal chip representation, and (iv) global classifier training with simultaneous autoencoder fine-tuning to achieve robust gas classification across multiple chips and temperature conditions.

The key contributions of this thesis are:

- A disentangled autoencoder architecture that decouples temperature compensation and fabrication error compensation, where both autoencoders are trained independently on normalized data, enabling independent optimization of each compensation objective while preventing error propagation inherent in sequential cascaded architectures.
- A joint training strategy that trains a global classifier from scratch while simultaneously fine-tuning the pre-trained autoencoders, allowing the compensation networks to adapt their representations to maximize classification performance.
- Comprehensive experimental evaluation demonstrating that the proposed method significantly outperforms baseline techniques while maintaining excellent generalization.
- Analysis showing that the proposed method effectively handles both temperature variations (25–29°C) and fabrication-induced spectral distortions, enabling robust cross-chip deployment without requiring chip-specific model retraining.

1.1 Thesis Organization

The remainder of this thesis is organized as follows:

Chapter 2 provides background on artificial neural networks, deep learning architectures including multilayer perceptrons, convolutional neural networks, and autoencoders. It also reviews data normalization techniques and discusses related work in spectroscopic analysis and photonic sensing.

Chapter 3 presents the proposed sequential autoencoder methodology in detail. It describes the four-stage pipeline including reference-based normalization, temperature compensation, fabrication error compensation, and global classifier training with joint fine-tuning. The chapter provides mathematical formulations for each component and

explains the training procedures.

Chapter 4 presents a comprehensive experimental evaluation of the proposed method. It describes the 10-chip dataset with fabrication errors and temperature variations, presents results for the proposed disentangled autoencoder pipeline, and conducts extensive ablation studies including baseline comparisons, architectural variants, and photonic hardware compatibility assessment.

Chapter 5 summarizes the key findings, discusses the implications for practical deployment of photonic gas sensors, and identifies promising directions for future research including extension to additional gas species, investigation of unsupervised autoencoder training, and adaptation to other photonic sensing modalities.

2

Background and Related Work

2.1 Background

2.1.1 Artificial Neural Networks

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain. They consist of interconnected processing units called **neurons**, which are organized in layers: an input layer, one or more hidden layers, and an output layer. Each neuron receives input signals, multiplies them by associated **weights**, adds a bias term, and applies a nonlinear **activation function** to produce an output. Mathematically, the operation of a neuron can be expressed as:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

where x_i are the input features, w_i the corresponding weights, b the bias term, $f(\cdot)$ the activation function (such as ReLU, sigmoid, or tanh), and n the total number of input features connected to the neuron.

2.1.2 Training of Neural Networks

The process of training a neural network involves adjusting its weights and biases so that the predicted outputs closely match the true targets. This is achieved through an iterative optimization process known as **backpropagation**. The steps are:

1. **Forward pass:** The input data are propagated through the network to produce predictions.

2. **Loss computation:** The difference between predicted and actual values is measured using a **loss function**, such as Mean Squared Error (MSE) or Cross-Entropy Loss.

3. **Backward pass:** The gradients of the loss with respect to each parameter are computed using the chain rule.

4. **Parameter update:** The network parameters are updated using an optimization algorithm such as Stochastic Gradient Descent (SGD) or Adam.

This cycle repeats for multiple **epochs**, allowing the model to minimize the loss and improve its predictive accuracy.

2.1.3 Deep Learning

Deep Learning refers to neural networks with multiple hidden layers capable of learning hierarchical and abstract representations of data. These models can automatically extract features from raw input, eliminating the need for manual feature engineering. Deep learning has shown remarkable success in domains such as image recognition, speech processing, and time-series analysis.

Various deep learning architectures have been developed to address different types of data and tasks. Multilayer Perceptrons (MLPs) serve as the foundation for fully-connected networks, effectively modeling complex nonlinear relationships in structured data. Convolutional Neural Networks (CNNs) extend these capabilities by incorporating spatial or temporal structure, enabling efficient feature extraction from sequential and grid-like data. Autoencoders provide unsupervised learning mechanisms for dimensionality reduction and denoising, learning compact representations that capture essential data characteristics while discarding noise. Each architecture offers distinct advantages depending on the nature of the input data and the learning objective, and they are often combined to create powerful hybrid models tailored to specific application domains.

2.1.4 Multilayer Perceptrons (MLPs)

A **Multilayer Perceptron (MLP)** is a type of feedforward neural network composed of multiple layers of neurons, where each neuron in one layer is fully connected to all neurons in the next. The network maps an input vector $x \in \mathbb{R}^n$ to an output $y \in \mathbb{R}^k$ through a sequence of linear transformations and nonlinear activations. Mathematically, an MLP with L layers can be expressed as:

$$y = f^{(L)}(x) = \sigma^{(L)}(W^{(L)}\,\sigma^{(L-1)}(W^{(L-1)}\cdots\sigma^{(1)}(W^{(1)}x + b^{(1)})\cdots + b^{(L-1)}) + b^{(L)}),$$

where $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector of layer l, and $\sigma^{(l)}(\cdot)$ denotes the activation function (e.g., ReLU or tanh). MLPs are particularly effective for modeling nonlinear relationships in structured or tabular data, such as spectral sensor readings used in environmental pollution detection.

2.1.5 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) extend the concept of neural networks by introducing *convolution operations*, which enable the extraction of local spatial or temporal features from structured data. Instead of fully connected layers, CNNs apply learnable filters (kernels) that move across the input, computing local weighted sums to detect characteristic patterns such as edges, peaks, or correlations.

Mathematically, the convolution between an input x and a kernel w is expressed as:

$$y(\mathbf{r}) = (x * w)(\mathbf{r}) = \sum_{\mathbf{s}} x(\mathbf{r} - \mathbf{s})\, w(\mathbf{s}),$$

where $y(\mathbf{r})$ is the resulting feature map, $\mathbf{r}$ and $\mathbf{s}$ denote spatial (or temporal) coordinates, $x(\mathbf{r})$ is the input signal, and $w(\mathbf{s})$ is the kernel. In discrete form for multidimensional data (e.g., images or spectra), this operation corresponds to sliding the kernel over the input and computing element-wise multiplications followed by summation.

After convolution, a nonlinear activation function $\sigma(\cdot)$ is applied to introduce nonlinearity:

$$y(\mathbf{r}) = \sigma\big((x * w)(\mathbf{r}) + b\big).$$

This general formulation encompasses 1D, 2D, and 3D convolutions used for temporal, spatial, and volumetric data, respectively. In this work, one-dimensional convolutions are employed to analyze spectral sequences, capturing correlations among neighboring wavelength peaks in the sensor response.

Training and Optimization

Training a deep neural network involves adjusting its internal parameters (weights and biases) so that the model can accurately map input data to the desired outputs. This

process is formulated as an optimization problem, where the objective is to minimize a loss function that quantifies the discrepancy between the model's predictions and the ground truth labels.

For classification tasks, the most commonly used objective is the **cross-entropy loss**, which measures the dissimilarity between the predicted class probabilities and the true class labels:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}), \tag{2.1}$$

where N is the total number of training samples, C is the number of classes, $y_{i,c}$ is the true label (typically one-hot encoded), and $\hat{y}_{i,c}$ is the predicted probability that sample i belongs to class c.

The network parameters are optimized using gradient-based methods such as Stochastic Gradient Descent (SGD) [Bot12] or adaptive optimization algorithms like Adam [KB15]. The backpropagation algorithm efficiently computes the gradients of the loss function with respect to the network parameters, enabling iterative updates that minimize the loss and progressively improve the model's predictive performance.

2.1.6 Autoencoders

Autoencoders are neural networks trained to reconstruct their inputs via a low-dimensional latent code, thereby learning compact representations useful for denoising, dimensionality reduction, and feature learning [HS06, VLBM08]. They consist of two main components: an **encoder** that compresses the input into a lower-dimensional latent representation, and a **decoder** that reconstructs the input from this compressed code.

Encoder–Decoder Architecture

Let the encoder be defined as $f_\theta : \mathbb{R}^n \to \mathbb{R}^m$ and the decoder as $g_\phi : \mathbb{R}^m \to \mathbb{R}^n$. The encoder maps the high-dimensional input $\mathbf{x} \in \mathbb{R}^n$ to a latent representation $\mathbf{z} \in \mathbb{R}^m$ (with $m < n$), while the decoder reconstructs the input as $\hat{\mathbf{x}} \in \mathbb{R}^n$ (Figure 2.1):

$$\mathbf{z} = f_\theta(\mathbf{x}) \quad \text{(Encoder)} \tag{2.2}$$

$$\hat{\mathbf{x}} = g_\phi(\mathbf{z}) \quad \text{(Decoder)} \tag{2.3}$$

where θ and ϕ are the trainable parameter vectors of the encoder and decoder networks, respectively.

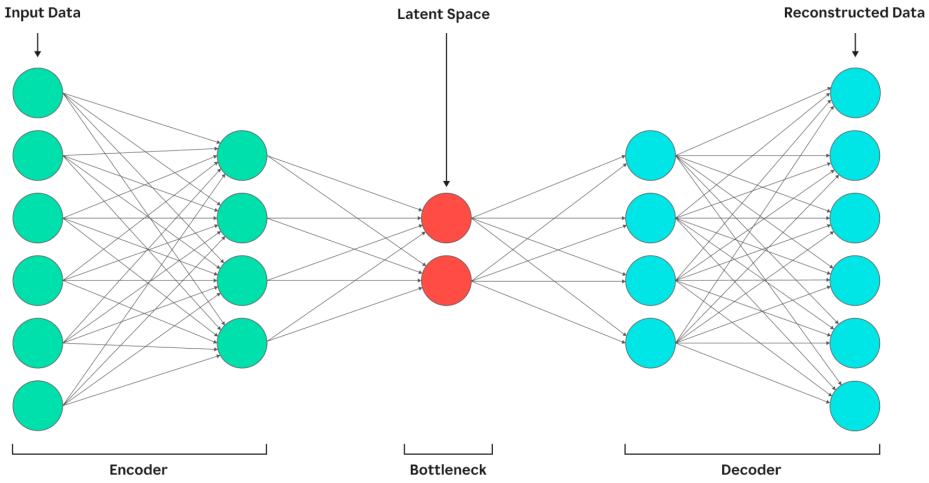


Figure 2.1: *Architecture of an autoencoder. The encoder compresses high-dimensional inputs to a latent bottleneck representation; the decoder reconstructs the input from this compressed code, encouraging the learning of compact, informative representations.*

The network is trained to minimize the reconstruction error between the input $\mathbf{x}_i$ and its reconstruction $\hat{\mathbf{x}}_i = g_\phi(f_\theta(\mathbf{x}_i))$:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 \tag{2.4}$$

where N is the number of training samples and $\|\cdot\|^2$ denotes the squared Euclidean norm. Here, $\hat{\mathbf{x}}_i$ represents the reconstructed version of the i-th input sample $\mathbf{x}_i$. Minimizing this loss encourages the network to learn latent representations that capture the essential structure of the data while discarding noise and irrelevant variations.

In sensor-based applications, autoencoders can learn device-invariant representations, mitigate noise and drift, and provide compact features for downstream classifiers.

2.1.7 Neural Network Components

This section describes the fundamental building blocks used in modern deep neural network architectures, including activation functions, normalization techniques, regularization methods, and pooling operations. These components are essential for understanding the architectures employed in this work.

Activation Functions

Activation functions introduce nonlinearity into neural networks, enabling them to learn complex patterns and relationships in data. Without activation functions, neural networks would be limited to learning linear transformations regardless of depth.

Sigmoid. The sigmoid activation function is one of the earliest nonlinear activations used in neural networks. It is defined as:

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

The sigmoid function maps inputs to the range $(0, 1)$, making it suitable for binary classification tasks where outputs represent probabilities. However, it suffers from vanishing gradient problems for large positive or negative inputs, which can slow down training in deep networks.

Tanh (Hyperbolic Tangent). The hyperbolic tangent activation function is a scaled and shifted version of the sigmoid. It is defined as:

$$\text{Tanh}(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

Tanh maps inputs to the range $(-1, 1)$, providing zero-centered outputs that often lead to faster convergence compared to sigmoid. Like sigmoid, tanh can also suffer from vanishing gradients for extreme input values.

ReLU (Rectified Linear Unit). The Rectified Linear Unit is one of the most widely used activation functions in modern deep learning. It is defined as:

$$\text{ReLU}(x) = \max(0, x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

ReLU addresses the vanishing gradient problem by providing a constant gradient of 1 for positive inputs, enabling efficient training of deep networks. However, it can suffer from the ”dying ReLU” problem where neurons output zero for all inputs if their weights are updated such that they always receive negative inputs.

GELU (Gaussian Error Linear Unit). The Gaussian Error Linear Unit (GELU) is a smooth activation function commonly used in modern deep learning architectures. GELU is defined as:

$$\text{GELU}(x) = x \cdot \Phi(x) = x \cdot \frac{1}{2}\left[1 + \text{erf}\left(\frac{x}{\sqrt{2}}\right)\right]$$

where $\Phi(x)$ is the cumulative distribution function of the standard Gaussian distribution. GELU provides smooth, non-monotonic activation with probabilistic behavior, enabling better gradient flow and feature learning compared to ReLU. The smooth nature of GELU helps prevent gradient vanishing and allows for more stable optimization in deep networks.

ELU (Exponential Linear Unit). The Exponential Linear Unit (ELU) is an activation function particularly effective for convolutional neural networks. ELU is defined as:

$$\text{ELU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(e^x - 1) & \text{if } x \leq 0 \end{cases}, \quad \alpha = 1.0$$

ELU provides smooth negative values that help with gradient flow and robustness to noise in spatial feature extraction. Unlike ReLU, ELU outputs can take negative values, which helps push mean activations closer to zero and accelerates learning.

Softmax. The softmax activation function converts a vector of real-valued scores (logits) into a probability distribution over multiple classes. For a vector $\mathbf{z} \in \mathbb{R}^C$ of logits, the softmax function is defined as:

$$\text{Softmax}(\mathbf{z})_i = \frac{\exp(z_i)}{\sum_{j=1}^{C} \exp(z_j)}, \quad i = 1, \ldots, C$$

where C is the number of classes. Softmax ensures that all output values lie in $(0, 1)$ and sum to 1, making them interpretable as class probabilities. It is commonly used in the final layer of classification networks, where it transforms raw network outputs into normalized probability distributions suitable for multi-class decision making.

Batch Normalization

Batch normalization is a technique that normalizes the inputs of each layer by adjusting and scaling activations. It stabilizes training by reducing internal covariate shift and enables higher learning rates. Batch normalization is applied immediately after linear or

convolutional transformations and before activation functions.

The general batch normalization operation can be expressed as:

$$\text{BN}(\mathbf{h}) = \gamma \frac{\mathbf{h} - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} + \beta$$

where $\mu_{\mathcal{B}}$ and $\sigma_{\mathcal{B}}^2$ are the batch mean and variance, γ is a learnable scale parameter and β is a learnable shift parameter, and ϵ is a small constant for numerical stability (typically 10^{-5}). During inference, batch statistics are replaced with running estimates computed during training.

In convolutional neural networks, batch normalization operates across the channel dimension:

$$\text{BatchNorm1d}(\mathbf{C}) = \gamma \frac{\mathbf{C} - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} + \beta$$

where $\mathbf{C} \in \mathbb{R}^{C_{\text{out}} \times L}$ represents the feature maps, and normalization is applied independently to each channel across the batch and spatial dimensions.

Dropout Regularization

Dropout is a regularization technique that randomly sets a fraction of neuron activations to zero during training, preventing co-adaptation of neurons and improving generalization. During training, each neuron's activation is randomly set to zero with probability p (the dropout rate) and scaled by $\frac{1}{1-p}$ during inference to maintain expected activation magnitudes.

Pooling Operations

Pooling operations are downsampling techniques that reduce the spatial or temporal dimensionality of feature representations while preserving essential information. By aggregating local neighborhoods of features, pooling improves computational efficiency, provides robustness to small spatial variations, and helps prevent overfitting. Two commonly used pooling strategies are max pooling and average pooling:

Max Pooling. Max pooling extracts the maximum value from each local region of the input, effectively selecting the most prominent feature within that region. This operation preserves the strongest activations while discarding weaker responses, making it particularly useful for detecting salient patterns regardless of their exact position. Max pooling can be applied with fixed window sizes or adaptively adjusted to produce a

desired output dimensionality, enabling flexible dimensionality reduction in hierarchical architectures.

Average Pooling. Average pooling computes the mean of values within each local region, providing a smoother aggregation of features compared to max pooling. This approach captures the overall presence of features rather than focusing solely on peak activations. When applied globally across entire feature maps, average pooling produces fixed-size representations regardless of input dimensions, making it suitable for transitioning from convolutional feature extraction to fully-connected classification layers.

2.1.8 Loss Functions and Training Objectives

This section describes common loss functions used to train neural networks for different tasks. Two fundamental objectives are reconstruction and classification.

Mean Squared Error Loss

Mean squared error (MSE) loss quantifies the average squared difference between predicted and target values. While commonly used for reconstruction tasks in autoencoders, MSE is applicable to any regression problem where the goal is to minimize prediction error:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2^2$$

where N is the batch size, $\mathbf{y}_i$ is the target, and $\hat{\mathbf{y}}_i$ is the predicted output. The squared Euclidean norm $\|\cdot\|_2^2$ heavily penalizes large errors, making MSE sensitive to outliers but effective for learning accurate continuous predictions. In autoencoders, MSE measures reconstruction fidelity by comparing inputs $\mathbf{x}_i$ with their reconstructions $\hat{\mathbf{x}}_i$, encouraging the network to preserve essential information while discarding noise.

Cross-Entropy Loss

Cross-entropy loss measures the discrepancy between predicted class probabilities and true labels, widely used for training classification models:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c})$$

where N is the batch size, $y_{i,c}$ is the true label (1 if sample i belongs to class c, 0 otherwise), $\hat{y}_{i,c}$ is the predicted probability for class c obtained by applying the softmax activation function to the network's logits, and C is the number of classes. The negative log-likelihood penalizes confident incorrect predictions, encouraging the model to assign high probabilities to the correct classes.

2.1.9 Data Normalization in Machine Learning

Data normalization is a critical preprocessing step that ensures features contribute proportionally to learning, independent of their original scales or units. Without normalization, large-magnitude features can dominate optimization, bias model behavior, and slow convergence [LBOM12]. Normalization improves numerical stability, aids generalization, and enables efficient gradient-based optimization—especially in deep networks where input magnitudes influence training dynamics.

In the context of spectroscopic and sensor-based data, normalization is essential to compensate for variations in signal intensity, environmental conditions, or device response. This work employs a two-stage normalization strategy: **robust scaling with reference-class anchoring** as the primary normalization step, followed by an optional **global standardization** using Z-score normalization to further harmonize features across all chips.

Robust Scaling with Reference-Class Anchoring

Robust scaling is a normalization technique that uses median and Median Absolute Deviation (MAD) statistics to reduce sensitivity to outliers. Given a data sample $\mathbf{x} \in \mathbb{R}^M$ and a reference distribution with median vector $\mathbf{m}_{\text{ref}}$, the normalized sample $\tilde{\mathbf{x}}$ is computed as

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mathbf{m}_{\text{ref}}}{\mathbf{d}}, \tag{2.5}$$

where $\mathbf{d}$ is the Median Absolute Deviation (MAD) vector, defined as

$$\mathbf{d} = \text{median}\big(\{\, |\mathbf{x}_i^{\text{ref}} - \mathbf{m}_{\text{ref}}| \,\}_{i=1}^{N_{\text{ref}}}\big), \tag{2.6}$$

with N_{ref} denoting the number of reference-class samples and i serving as an index over them.

By centering each sample at the median $\mathbf{m}_{\text{ref}}$ and scaling by the MAD-based dispersion $\mathbf{d}$, this method provides robust normalization that is insensitive to extreme values and outliers, making it particularly suitable for noisy or skewed data distributions. The use

of median and MAD statistics ensures stable estimation even when the data contains anomalies or non-Gaussian characteristics.

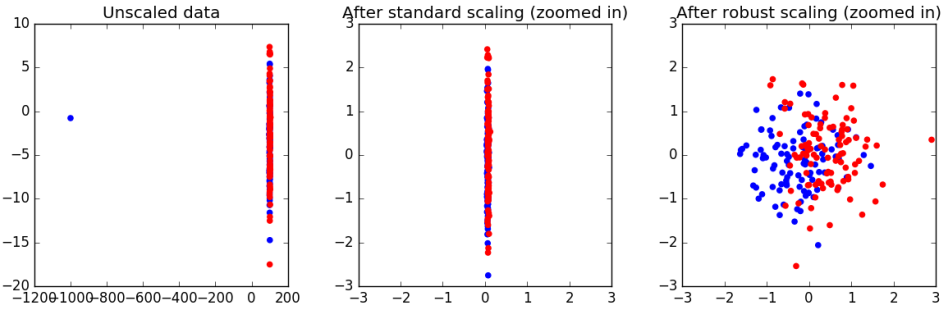


Figure 2.2: *Median–MAD normalization reduces sensitivity to outliers, yielding more stable feature distributions for noisy data.*

Global Standardization (Optional)

Following robust scaling, an optional global standardization step can be applied to further harmonize feature distributions across the entire dataset. This secondary normalization uses Z-score standardization, which centers each feature at zero mean and scales it to unit variance. For a feature vector $\mathbf{x}' \in \mathbb{R}^M$ already normalized by robust scaling, the global standardization applies:

$$\hat{x}_j = \frac{x'_j - \mu_j^{\text{global}}}{\sigma_j^{\text{global}} + \varepsilon}, \qquad j = 1, \ldots, M, \tag{2.7}$$

where μ_j^{global} and σ_j^{global} are the mean and standard deviation of feature j computed from all training samples in the dataset, and $\varepsilon > 0$ is a small constant for numerical stability. This global standardization ensures that all features have comparable scales and distributions after the initial robust normalization, facilitating more effective learning. The combined two-stage normalization strategy leverages the complementary strengths of robust statistics (outlier resistance) and global standardization (feature harmonization).

2.2 Related Work

Environmental pollution detection has become a critical research domain due to the growing demand for continuous and distributed air-quality monitoring. Traditional gas detection systems, such as optical benches, electrochemical sensors, or infrared absorption analyzers, though highly accurate, are often bulky, expensive, and unsuitable for large-scale or portable deployments. In recent years, deep learning has emerged as a powerful

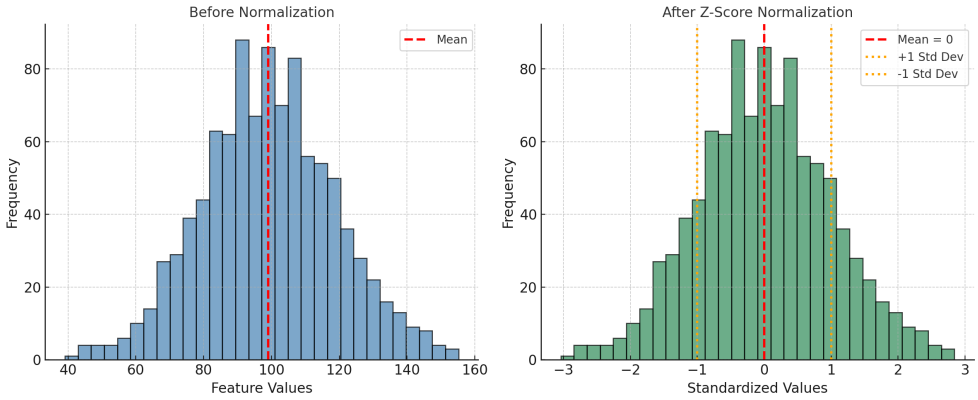


Figure 2.3: *Global Z-score standardization applied after robust scaling centers each feature across all chips and scales it to unit variance, ensuring consistent feature magnitudes for cross-chip learning.*

tool for environmental applications, enabling automatic feature extraction and pattern recognition in complex, multidimensional data. Convolutional and recurrent architectures have been successfully applied to detect pollution sources from hyperspectral imagery [DS20] and to predict pollutant concentrations over time using distributed sensor networks [LZ21]. These advances highlight the potential of data-driven methods to model diverse environmental phenomena and improve the reliability of pollution monitoring systems.

Motivated by these successes, research has increasingly focused on integrating deep learning with optical spectrometry for gas detection. Spectroscopic sensing provides direct access to molecular absorption features, allowing precise identification of gases such as CO_2, CH_4, and NH_3. However, traditional optical setups based on bulk optics require complex alignment, large physical footprints, and high power consumption, making them impractical for scalable deployment in distributed sensor networks [VCB+13]. The advent of integrated photonics has addressed these limitations, enabling compact, low-cost, and energy-efficient on-chip spectrometers capable of real-time spectral analysis in uncontrolled environments [LF21, VCB+13]. Within this context, deep learning models have been explored to classify or reconstruct spectral responses, enhancing sensitivity and robustness against measurement noise.

Among integrated photonic architectures, the Spatial Heterodyne Fourier Transform (SHFT) spectrometer has emerged as a particularly promising solution for miniaturized gas detection [HBLK+19a]. The SHFT design leverages arrays of Mach–Zehnder Interferometers (MZIs) to encode spectral information into interferometric patterns that can be processed by machine learning algorithms. Previous studies have shown that supervised deep learning models can accurately classify interferograms corresponding to different gas absorption signatures [KAR+25, HBLK+19b]. Nevertheless, these models typically assume stable device characteristics and operating conditions, leading to

degraded performance when applied across chips or under varying temperatures due to fabrication tolerances and thermal drifts.

To overcome these challenges, this work proposes a purely data-driven compensation framework based on sequential autoencoders. The proposed pipeline explicitly learns to correct fabrication- and temperature-induced distortions in the measured interferometric responses from data alone, without requiring detailed physical models of the device characteristics. The method consists of four key stages: (i) a reference-based normalization step that anchors each chip to its no-gas reference statistics, (ii) a temperature compensation autoencoder that maps measurements acquired at arbitrary temperatures to a fixed reference temperature (iii) a fabrication error compensation autoencoder that aligns chip-specific responses to a common ideal chip representation, and (iv) global classifier training with simultaneous autoencoder fine-tuning to achieve robust gas classification across chips and conditions. The proposed design focuses on improving representation stability and cross-device generalization, as detailed in Chapter 3.

3

Proposed Method

In this work, we propose a comprehensive training pipeline that compensates for fluctuations caused by fabrication errors and temperature shifts using sequential autoencoders, aiming to improve the performance of gas absorption classification models on Spatial Heterodyne Fourier-Transform (SHFT) spectrometers. The proposed methodology consists of four distinct stages: (a) reference-based normalization, (b) temperature compensation through a temperature compensation autoencoder, (c) fabrication error compensation through a fabrication error compensation autoencoder, and (d) global classifier training with simultaneous autoencoder fine-tuning for final gas classification. Each stage of the proposed pipeline is described in detail below.

3.1 Reference-based Normalization

During the data collection phase, a dataset, denoted as $\mathcal{X}$, is constructed, consisting of samples $\{\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^M\}$, where M represents the number of Mach-Zehnder interferometers (MZIs). Each sample $\boldsymbol{x}$ represents the raw interferometric response obtained under different environmental and operational conditions from SHFT spectrometers and is labeled with a label $l \in \{1, ..., C\}$, corresponding to one of the C classes. Specifically, the dataset contains samples of $C - 1$ gas classes and one reference class that represents the absence of gases, serving as a baseline for normalization. Due to fluctuations in temperature and fabrication tolerances, the collected data contain distortions and noise that deviate from the ideal spectrometer responses, which can negatively affect the performance of the classification models.

To ensure consistency across measurements and to emphasize the variations introduced by each gas, a reference-based normalization is applied as a preprocessing step. Specifically, the responses of the different samples, representing different gas classes, are normalized with respect to the statistics of the reference class, effectively capturing only the

differential features between the gas-exposed and reference conditions. This process reduces the influence of baseline variations and highlights the differential features introduced by the gas, providing more stable and discriminative input data for subsequent autoencoder and classification stages.

For robust scaler-based normalization, let $\mathbf{x}$ denote the interferometric response corresponding to a specific gas class, while $\mathbf{m}_{\text{ref}}$ represents the median vector of the interferometric responses of the reference class. The normalized interferometric response $\tilde{\mathbf{x}}$ is then calculated as:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mathbf{m}_{\text{ref}}}{\mathbf{d}}, \tag{3.1}$$

where $\mathbf{d}$ is the Median Absolute Deviation (MAD) vector, which is computed as the median of the absolute differences between the reference interferometric responses and the median vector $\mathbf{m}_{\text{ref}}$:

$$\mathbf{d} = \text{median}\left(\left\{\left|\mathbf{x}_i^{\text{ref}} - \mathbf{m}_{\text{ref}}\right|\right\}_{i=1}^{N_{\text{ref}}}\right), \tag{3.2}$$

where N_{ref} is the number of samples in the reference class. This normalization procedure standardizes each interferometric measurement with respect to the statistical characteristics of the reference class, effectively reducing baseline variations and emphasizing the differential features introduced by the gas.

3.2 Temperature Compensation Autoencoder

Although the normalized spectral responses emphasize gas-specific features, they still retain temperature-induced distortions. To address these thermal variations, a temperature compensation autoencoder is employed to reconstruct the distorted interferometric representations acquired at arbitrary temperatures into their corresponding clean representations at a fixed temperature T_{ref}.

3.2.1 Data Preparation for Temperature Compensation

Following normalization using the robust scaler method, the data undergoes a preparation stage to facilitate training of the temperature compensation autoencoder. For each chip, measurements are acquired at various temperatures. To enable the autoencoder to learn a mapping from arbitrary temperature conditions to a standardized fixed temperature

T_{ref}, we create matched training pairs. For each gas class c, a class-representative sample at T_{ref} is computed by averaging all samples of class c measured at T_{ref} (within a specified tolerance):

$$\mathbf{x}_c^{\text{ref}} = \frac{1}{N_c^{\text{ref}}} \sum_{i=1}^{N_c^{\text{ref}}} \mathbf{x}_i^{c,\text{ref}}, \tag{3.3}$$

where N_c^{ref} denotes the number of such samples. Each training sample $\mathbf{x}_j$ measured at any temperature T_j is then paired with its corresponding class-representative target $\mathbf{x}_{c_j}^{\text{ref}}$, where c_j is the class label of sample j. The autoencoder input includes both the M spectral features and the temperature value T_j as an additional feature (yielding $M+1$ input features), while the target $\mathbf{x}_{c_j}^{\text{ref}}$ contains only the M spectral features at the fixed temperature. This matching strategy is applied independently to each chip, accounting for chip-specific fabrication characteristics.

3.2.2 Autoencoder Architecture and Training

Let $f_{\text{enc}}^{(1)}(\cdot)$ and $f_{\text{dec}}^{(1)}(\cdot)$ denote the encoder and decoder functions of the first autoencoder, respectively. Given a normalized interferometric response $\tilde{\mathbf{x}}$ with temperature information concatenated as $[\tilde{\mathbf{x}}, T] \in \mathbb{R}^{M+1}$, the encoder projects it into a latent representation:

$$\mathbf{z}^{(1)} = f_{\text{enc}}^{(1)}([\tilde{\mathbf{x}}, T]), \tag{3.4}$$

which is then reconstructed by the decoder as:

$$\mathbf{x}^{\text{rec},(1)} = f_{\text{dec}}^{(1)}(\mathbf{z}^{(1)}), \tag{3.5}$$

where $\mathbf{x}^{\text{rec},(1)} \in \mathbb{R}^M$ represents the temperature-compensated spectral response, corresponding to the reconstructed response at the fixed temperature T_{ref}.

To ensure that the reconstructed representations maintain the discriminative features necessary for gas classification, a classification head is incorporated in addition to the encoder-decoder structure. While the encoder-decoder structure removes temperature-induced noise, the addition of the classification head enforces that the latent representations preserve the information required to distinguish between gas classes. Without this auxiliary objective, the autoencoder could produce clean reconstructions that are close to the ideal in terms of temperature normalization but lose the class-discriminative differences that are critical for accurate classification. By simultaneously optimizing the reconstruction and classification losses, the network generates representations that are both temperature-invariant and class-discriminative,

making them suitable for downstream processing.

Let $f_{\text{cls}}^{(1)}(\cdot)$ denote the classification head attached to the first encoder. Given the latent representation $\mathbf{z}^{(1)}$, the classification head produces a class prediction:

$$\mathbf{y}^{(1)} = f_{\text{cls}}^{(1)}(\mathbf{z}^{(1)}). \tag{3.6}$$

The entire temperature compensation autoencoder network is trained end-to-end by minimizing a combined loss that balances reconstruction fidelity and classification accuracy:

$$\mathcal{L}_{\text{total}}^{(1)} = \lambda_{\text{rec}}^{(1)}\mathcal{L}_{\text{rec}}^{(1)} + \lambda_{\text{cls}}^{(1)}\mathcal{L}_{\text{cls}}^{(1)}, \tag{3.7}$$

where $\lambda_{\text{rec}}^{(1)}$ and $\lambda_{\text{cls}}^{(1)}$ are weighting coefficients that balance the two objectives. The reconstruction loss $\mathcal{L}_{\text{rec}}^{(1)}$ measures the similarity between the reconstructed and target temperature-normalized interferometric responses using the Mean Squared Error (MSE) metric:

$$\mathcal{L}_{\text{rec}}^{(1)} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_i^{\text{rec},(1)} - \mathbf{x}_{c_i}^{\text{ref}}\right\|_2^2, \tag{3.8}$$

where N is the number of training samples, and $\mathbf{x}_{c_i}^{\text{ref}}$ is the class-representative target at the reference temperature T_{ref} for sample i.

The classification loss $\mathcal{L}_{\text{cls}}^{(1)}$ enforces discriminative learning in the latent space, expressed as the cross-entropy loss:

$$\mathcal{L}_{\text{cls}}^{(1)} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} l_{i,c}\log(\hat{y}_{i,c}^{(1)}), \tag{3.9}$$

where $\hat{y}_{i,c}^{(1)} = f_{\text{cls}}^{(1)}(f_{\text{enc}}^{(1)}([\tilde{\mathbf{x}}_i, T_i]))$ is the predicted probability for class c, and $l_{i,c}$ is the corresponding ground-truth label.

This combined optimization enables the autoencoder to produce temperature-normalized, denoised representations that retain class-discriminative features. The training is performed independently for each chip, resulting in chip-specific autoencoders that account for the unique temperature response characteristics of each device.

3.3 Fabrication Error Compensation Autoencoder

While the temperature compensation autoencoder compensates for temperature variations, the spectral responses still contain chip-specific distortions caused by fabrication imperfections. Different chips exhibit unique systematic deviations from the ideal spectrometer response due to variations in the manufacturing process. To address these chip-to-chip variations and enable cross-chip generalization, a fabrication error compensation autoencoder is employed to map chip-specific spectral representations to a universal ideal chip representation.

An ideal chip representation, serving as the target for fabrication error compensation, is defined to provide the standardized spectral space. This ideal representation is assumed to exhibit minimal fabrication errors and provide high-quality spectral responses across all gas classes and temperature conditions. Each chip has its own fabrication error compensation autoencoder that learns to transform the normalized spectral responses from that specific chip into the spectral space of the ideal chip, effectively standardizing the representations and removing chip-specific fabrication-induced discrepancies.

It is important to note that the training of the fabrication error compensation autoencoder is performed independently from the temperature compensation autoencoder. Specifically, both autoencoders take as input the normalized chip spectral data (output of the normalization stage, $\tilde{\mathbf{x}} \in \mathbb{R}^M$). The fabrication error compensation autoencoder learns to map it directly to the corresponding ideal chip representation, matched by both class and temperature. This independent training strategy allows the fabrication error compensation autoencoder to learn fabrication error compensation without being affected by potential imperfections in the temperature compensation autoencoder's outputs.

3.3.1 Data Preparation for Fabrication Error Compensation

Following normalization using the robust scaler method, the data undergoes a preparation stage to facilitate training of the fabrication error compensation autoencoder. For each chip, normalized spectral measurements are prepared as inputs to the autoencoder. To enable the autoencoder to learn a mapping from chip-specific fabrication characteristics to a standardized ideal chip representation, we create matched training pairs by aligning each chip sample with the corresponding ideal chip sample based on both class label and temperature.

For each training sample $\mathbf{x}_j$ from a given chip with class label c_j and temperature T_j, we identify the corresponding ideal chip target by matching both attributes. Specifically, we search the normalized ideal chip dataset for samples where the class label matches c_j and

the temperature value matches T_j (within a specified tolerance ϵ):

$$\mathbf{x}_{c_j,T_j}^{\text{ideal}} = \text{ideal_chip_sample}(c = c_j, |T - T_j| \leq \epsilon), \tag{3.10}$$

where ϵ is a small temperature tolerance (typically 0.03 °C) to account for discretization in the temperature schedule. This dual-matching strategy ensures that the target ideal chip representation preserves both the class-specific spectral signatures and the temperature-dependent characteristics inherent in the original sample. In cases where multiple ideal chip samples match the criteria, we select the first match; if no exact match is found within the tolerance, we fall back to class-only matching, and as a last resort, use a zero-filled vector.

The autoencoder input consists of the M normalized spectral features $\tilde{\mathbf{x}}_j \in \mathbb{R}^M$ from the given chip, while the target $\mathbf{x}_{c_j,T_j}^{\text{ideal}} \in \mathbb{R}^M$ contains the corresponding M spectral features from the ideal chip. This matching strategy is applied independently to each chip, allowing each chip's fabrication error compensation autoencoder to learn a specialized mapping from its unique fabrication-induced distortions to the universal ideal chip representation.

3.3.2 Autoencoder Architecture and Training

Let $f_{\text{enc}}^{(2)}(\cdot)$ and $f_{\text{dec}}^{(2)}(\cdot)$ denote the encoder and decoder functions of the second autoencoder, respectively. Given a normalized chip spectral response $\tilde{\mathbf{x}}$, the encoder projects it into a latent representation:

$$\mathbf{z}^{(2)} = f_{\text{enc}}^{(2)}(\tilde{\mathbf{x}}), \tag{3.11}$$

which is then reconstructed by the decoder as:

$$\mathbf{x}^{\text{rec},(2)} = f_{\text{dec}}^{(2)}(\mathbf{z}^{(2)}), \tag{3.12}$$

where $\mathbf{x}^{\text{rec},(2)} \in \mathbb{R}^M$ represents the transferred spectral response in the ideal chip space.

For training the fabrication error compensation autoencoder, we construct ideal target representations $\mathbf{x}_{c,T}^{\text{ideal}}$ from the full normalized ideal chip dataset by matching each chip sample with the corresponding ideal chip sample of the same class c and temperature T. This matching process inherently preserves both class-specific spectral signatures and temperature-dependent characteristics in the target space. However, to ensure that these discriminative features are explicitly retained in the learned latent representations, we incorporate a classification head during training. This auxiliary classification objective

enforces that the encoder extracts class-discriminative features into the latent space $\mathbf{z}^{(2)}$, preventing the autoencoder from learning degenerate solutions that achieve low reconstruction error but lose class separability. Without this explicit supervision in the latent space, the autoencoder could potentially learn to map different classes to overlapping regions, relying solely on the decoder to reconstruct the correct class-specific patterns. By adding the classification head, we ensure that the latent representations themselves are class-discriminative, which is crucial for the subsequent global classifier training stage where these representations serve as input features.

Similar to the temperature compensation autoencoder, the fabrication error compensation autoencoder incorporates a classification head $f_{\text{cls}}^{(2)}(\cdot)$ to preserve class discriminability:

$$\mathbf{y}^{(2)} = f_{\text{cls}}^{(2)}(\mathbf{z}^{(2)}). \tag{3.13}$$

The fabrication error compensation autoencoder is trained by minimizing a combined loss:

$$\mathcal{L}_{\text{total}}^{(2)} = \lambda_{\text{rec}}^{(2)}\mathcal{L}_{\text{rec}}^{(2)} + \lambda_{\text{cls}}^{(2)}\mathcal{L}_{\text{cls}}^{(2)}, \tag{3.14}$$

where the reconstruction loss is:

$$\mathcal{L}_{\text{rec}}^{(2)} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_i^{\text{rec},(2)} - \mathbf{x}_{c_i,T_i}^{\text{ideal}}\right\|_2^2, \tag{3.15}$$

and the classification loss is:

$$\mathcal{L}_{\text{cls}}^{(2)} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} l_{i,c}\log(\hat{y}_{i,c}^{(2)}). \tag{3.16}$$

This independent training approach offers several advantages: (1) it prevents error accumulation that could occur if the fabrication error compensation autoencoder were trained sequentially on potentially imperfect outputs from the temperature compensation autoencoder, and (2) it enables independent optimization of temperature and fabrication compensation objectives, as each autoencoder learns directly from the normalized chip data rather than from the output of the other. The training is performed independently for each chip, resulting in chip-specific fabrication error compensation autoencoders that learn to map each chip's unique fabrication characteristics to the ideal representation.

3.4 Global Classifier Training with Autoencoder Fine-Tuning

In the final stage of the proposed pipeline, we train a global classifier while simultaneously fine-tuning both pre-trained autoencoders to achieve optimal gas classification performance across all chips. Unlike the previous stages where each chip was processed independently, this stage integrates data from all chips to train a unified classification model.

The training procedure operates as follows. First, data from all chips are loaded and split into training, validation, and test sets using a stratified split to ensure balanced class representation across all subsets. For each chip k, we load the corresponding pre-trained temperature compensation autoencoder $(f^{(1)}_{\text{enc},k}, f^{(1)}_{\text{dec},k})$ and fabrication error compensation autoencoder $(f^{(2)}_{\text{enc},k}, f^{(2)}_{\text{dec},k})$. These autoencoders are initialized with their pre-trained weights obtained from the independent training stages, and their parameters remain trainable during this stage, enabling fine-tuning simultaneously with the global classifier training.

For a given input sample $\mathbf{x}_j$ from chip k with temperature T_j, the sequential forward pass through the pipeline is defined as:

$$\mathbf{x}_j^{\text{rec},(1)} = f^{(1)}_{\text{dec},k}(f^{(1)}_{\text{enc},k}([\tilde{\mathbf{x}}_j, T_j])), \tag{3.17}$$

$$\mathbf{x}_j^{\text{rec},(2)} = f^{(2)}_{\text{dec},k}(f^{(2)}_{\text{enc},k}(\mathbf{x}_j^{\text{rec},(1)})), \tag{3.18}$$

where $\mathbf{x}_j^{\text{rec},(2)} \in \mathbb{R}^M$ represents the final processed representation that is both temperature-compensated and fabrication-compensated.

A global classifier $g_{\text{cls}}(\cdot)$, shared across all chips, is then applied to this final representation to produce the gas class prediction:

$$\hat{\mathbf{y}}_j = g_{\text{cls}}(\mathbf{x}_j^{\text{rec},(2)}). \tag{3.19}$$

All model components—both autoencoders for all chips and the global classifier—are updated simultaneously by minimizing the global classification loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{global}} = -\frac{1}{N}\sum_{j=1}^{N}\sum_{c=1}^{C} l_{j,c}\log(\hat{y}_{j,c}), \tag{3.20}$$

where $\hat{y}_{j,c}$ is the predicted probability for class c from the global classifier, and $l_{j,c}$ is the corresponding ground-truth label. This objective allows the autoencoders to adapt their learned transformations directly based on the end-to-end classification task, ensuring that the final representations maximize class discriminability.

This approach offers several key advantages:

1. **End-to-end optimization:** By optimizing all components simultaneously through the global classification objective, the autoencoders can adapt their representations to maximize the performance of the global classifier, rather than being frozen with their independently-optimized weights.

2. **Cross-chip generalization:** The global classifier learns to recognize gas signatures in the unified ideal representation space, enabling robust classification across chips with different fabrication characteristics.

3. **Adaptive feature extraction:** The autoencoders fine-tune their learned transformations based on feedback from the classification task, ensuring that the most discriminative features for gas detection are preserved and enhanced.

After training, the final classification model consists of the chip-specific autoencoder pairs and the global classifier. At inference time, for a new sample $\mathbf{x}_{\text{new}}$ from chip k measured at temperature T_{new}, the prediction is obtained by sequentially passing the sample through the temperature compensation autoencoder, the fabrication error compensation autoencoder, and finally the global classifier:

$$\hat{l}_{\text{new}} = \arg\max_{c}\; g_{\text{cls}}\Big(f^{(2)}_{\text{dec},k}\Big(f^{(2)}_{\text{enc},k}\Big(f^{(1)}_{\text{dec},k}\Big(f^{(1)}_{\text{enc},k}([\tilde{\mathbf{x}}_{\text{new}}, T_{\text{new}}])\Big)\Big)\Big)\Big). \tag{3.21}$$

It is worth emphasizing that while the auxiliary classification heads in the autoencoders play a crucial role during the independent training stages (Stages 2 and 3) to ensure that discriminative features are preserved, they are not used during the global classifier training and fine-tuning stage (Stage 4), nor for the final gas detection decision. Instead, the global classifier, trained on the refined and standardized representations produced by the sequential autoencoders, serves as the sole decision-making component for gas classification. This design ensures that the autoencoders focus on their primary

objectives—temperature compensation and fabrication error compensation—while the global classifier leverages these optimized representations to achieve the highest prediction accuracy.

3.5 Pipeline Overview

The complete proposed pipeline is illustrated in Figure 3.1. The sequential processing stages transform raw interferometric measurements into robust, standardized representations suitable for accurate gas classification, even in the presence of significant temperature variations and fabrication errors. Experimental results demonstrate that the proposed pipeline significantly enhances detection accuracy and achieves robust performance under diverse temperature and fabrication conditions, as detailed in Chapter 4.

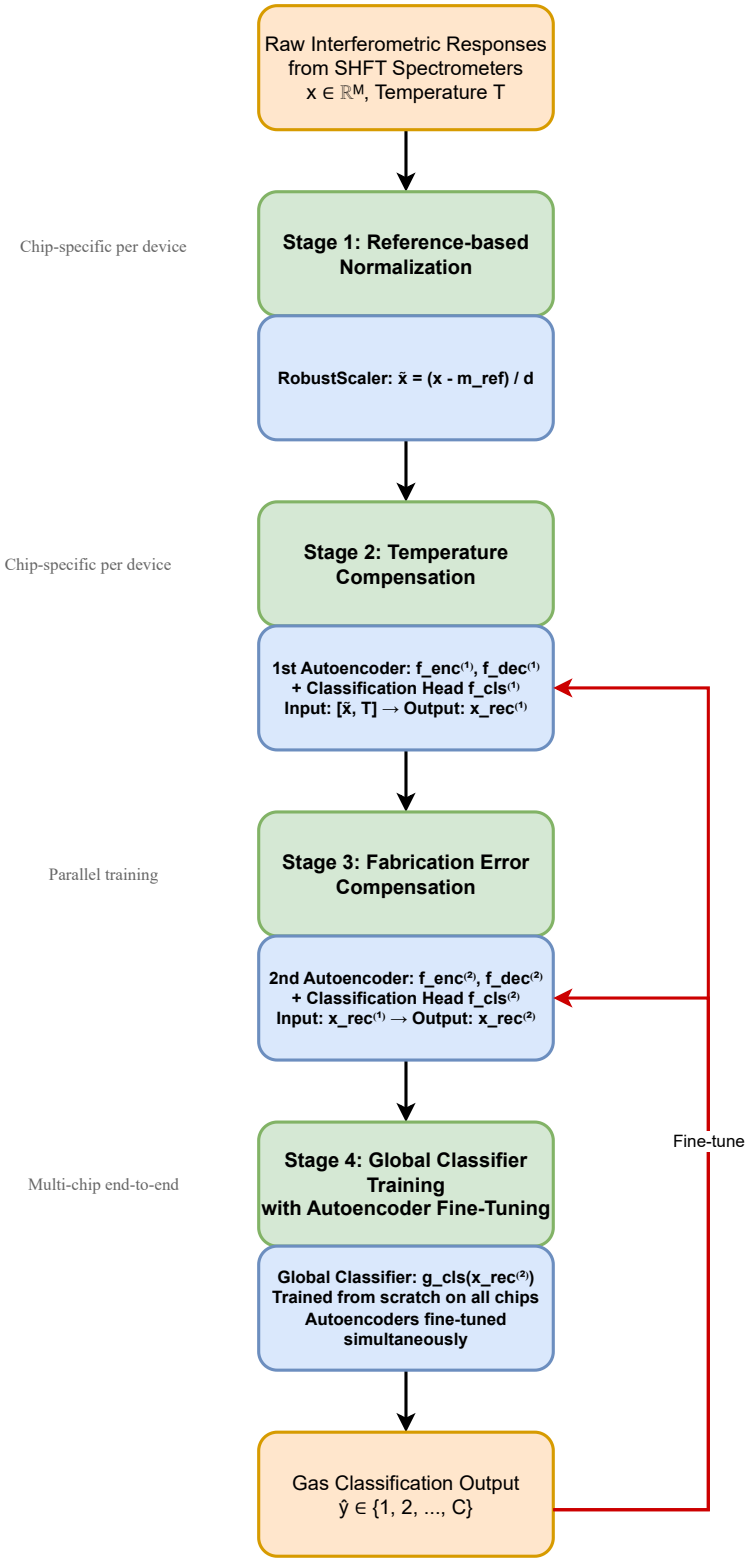


Figure 3.1: *Overview of the proposed pipeline for gas classification with sequential autoencoders. Raw interferometric responses from SHFT spectrometers undergo reference-based normalization, temperature compensation via the temperature compensation autoencoder, fabrication error compensation via the fabrication error compensation autoencoder, and finally global classifier training with simultaneous autoencoder fine-tuning for robust gas detection across multiple chips and temperature conditions.*

4

Experimental Evaluation

This chapter presents a comprehensive experimental evaluation of the proposed autoencoder-based framework for gas classification across multiple SHFT spectrometer chips under varying fabrication and temperature conditions. We begin by describing the dataset generation procedure and experimental setup in Section 4.1, followed by the neural network architectures employed in the framework in Section 4.2.

In Section 4.3, we present the experimental evaluation of the proposed method: a unified pipeline that integrates pre-trained temperature and fabrication compensation autoencoders with a global classifier trained on data from all chips. We also compare this proposed method against two baseline approaches that perform classification without autoencoders. Additionally, we validate the effectiveness of each autoencoder independently by training chip-specific classifiers on their outputs, demonstrating that each component learns its intended compensation mapping while the full proposed pipeline achieves superior performance.

Subsequently, we present additional comparative experiments to contextualize the results: raw data classification without preprocessing (Sections 4.4.1 and 4.4.2) and frozen autoencoders (Section 4.5). These experiments demonstrate the necessity of preprocessing and the advantages of the proposed autoencoder-based approach with fine-tuning.

4.1 Dataset Description

The dataset used in this study is generated in the interferometric domain from an on-chip Spatial Heterodyne Fourier Transform (SHFT) spectrometer comprising an array of Mach-Zehnder Interferometers (MZIs). Each chip model includes $M = 32$ MZIs with maximum optical path-length difference $\Delta L_{\max} = 1.13\,\text{cm}$, silicon core and silica cladding, and a nominal waveguide loss of $3\,\text{dB/cm}$. To emulate device variability, fabrication non-

idealities are injected as dimensional errors up to 20 nm (effective index in the range $n_{\text{eff}} \approx 2.04-2.20$) together with coupler-efficiency variations of $40\%-60\%$. Each sample is represented by the 32 interferometric outputs together with a temperature channel, yielding a $(M+1) = 33$-dimensional feature vector.

Classes and labeling. Four classes are considered: three spectra with a single absorption line at approximately 1549.8 nm, 1550.0 nm, and 1550.2 nm, and one reference class with no absorption line (no-gas condition). Labels correspond to the gas class (or absence of gas).

Temperature schedule. Thermal robustness is assessed by sweeping a dedicated temperature channel within each chip-class subset. For each subset we generate $N = 5000$ samples with a deterministic linear ramp from 25 °C to 29 °C:

$$T_i \;=\; 25 + \frac{4}{N-1}\,(i-1)\,^{\circ}\text{C}, \quad i = 1, \ldots, N,$$

so the per-sample increment is $\Delta T \approx 8.0 \times 10^{-4}\,^{\circ}\text{C}$.

Dataset size and splits. For a single chip: $4 \times 5000 = 20{,}000$ samples (33 features each). Experiments are repeated across 10 independent chip implementations; in aggregate the corpus contains 200,000 labeled samples. A stratified 70%/20%/10% train/validation/test split is applied to ensure balanced class representation across all subsets.

4.1.1 Summary of Key Parameters

- **Number of chips**: 10 (independent simulated implementations)
- **Number of classes**: 4 (three single-line absorptions at ∼1549.8/1550.0/1550.2 nm + one no-line reference)
- **Samples per class per chip**: 5000
- **Total samples**: 20,000 per chip; 200,000 in aggregate (10 chips)
- **MZIs per chip**: $M = 32$
- **Input dimensionality**: $M+1 = 33$ (32 interferometric outputs + temperature)

- **Temperature range**: Linear ramp 25 °C → 29 °C within each chip-class subset
- **Photonic parameters**: $\Delta L_{\max} = 1.13\,\text{cm}$; Si/SiO_2 waveguides; 3 dB/cm loss; fabrication errors up to 20 nm; coupler efficiency $40\% - 60\%$
- **Data splits**: 70% train / 20% validation / 10% test (stratified by class)

4.1.2 Evaluation Methodology

For each experimental configuration, we evaluate classification performance using the following metrics:

- **Accuracy**: Overall proportion of correctly classified samples.
- **Precision**: Per-class and macro-averaged precision, measuring the fraction of true positives among predicted positives.
- **Recall**: Per-class and macro-averaged recall, measuring the fraction of true positives among actual positives.
- **F1-score**: Harmonic mean of precision and recall, providing a balanced measure of classification performance.

All metrics are computed separately for training, validation, and test sets, with consistent metrics employed across all experiments to ensure comparability.

4.2 Models Architecture

This section describes the neural network architectures employed in the proposed framework: a denoising autoencoder for feature compensation (Section 4.2.1) and a convolutional classifier for gas classification (Section 4.2.2).

4.2.1 Linear Denoiser: Autoencoder Architecture

The `Linear Denoiser` is a fully-connected denoising autoencoder with an integrated classification head for multi-task learning. The architecture follows a symmetric encoder-decoder structure with a bottleneck latent representation. The model can be configured for two compensation tasks: temperature compensation (mapping measurements at arbitrary temperatures to a fixed reference temperature T_{ref}) and fabrication error compensation

(aligning chip-specific spectral characteristics to a unified ideal chip representation). The input consists of $d_{\text{in}} = 33$ features (32 interferometric measurements plus 1 temperature channel), compressed through an intermediate 64-dimensional representation to a 32-dimensional bottleneck $\mathbf{z}$, then reconstructed to $d_{\text{out}} = 32$ compensated measurements.

Encoder Architecture

The encoder consists of three fully-connected layers that progressively compress the input to a latent representation. The first layer transforms the input from 33 to 32 dimensions, followed by batch normalization, GELU activation, and 20% dropout. The second layer expands to an intermediate 64-dimensional representation with the same normalization, activation, and dropout scheme. The final bottleneck layer compresses to a 32-dimensional latent representation $\mathbf{z}$ using batch normalization and GELU activation without dropout to preserve latent information. The encoder forward pass is:

$$\mathbf{h}_1 = \text{Dropout}\big(\text{GELU}(\text{BN}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)), p = 0.2\big) \tag{4.1}$$

$$\mathbf{h}_2 = \text{Dropout}\big(\text{GELU}(\text{BN}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)), p = 0.2\big) \tag{4.2}$$

$$\mathbf{z} = \text{GELU}(\text{BN}(\mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3)) \tag{4.3}$$

where $\mathbf{x} \in \mathbb{R}^{33}$ is the input, $\mathbf{h}_1 \in \mathbb{R}^{32}$, $\mathbf{h}_2 \in \mathbb{R}^{64}$ is the intermediate representation, and $\mathbf{z} \in \mathbb{R}^{32}$ is the bottleneck representation used for classification.

Decoder Architecture

The decoder reconstructs the compensated output through symmetric expansion layers. The first layer expands the 32-dimensional bottleneck to 64 dimensions with batch normalization, GELU activation, and 20% dropout. The second layer further expands to 128 dimensions using the same processing scheme. The final output layer performs a linear transformation to reconstruct the 32-dimensional compensated measurements without activation, normalization, or dropout. The decoder forward pass is:

$$\mathbf{d}_1 = \text{Dropout}\big(\text{GELU}(\text{BN}(\mathbf{W}_4\mathbf{z} + \mathbf{b}_4)), p = 0.2\big) \tag{4.4}$$

$$\mathbf{d}_2 = \text{Dropout}\big(\text{GELU}(\text{BN}(\mathbf{W}_5\mathbf{d}_1 + \mathbf{b}_5)), p = 0.2\big) \tag{4.5}$$

$$\hat{\mathbf{x}} = \mathbf{W}_6\mathbf{d}_2 + \mathbf{b}_6 \tag{4.6}$$

where $\mathbf{z} \in \mathbb{R}^{32}$ is the bottleneck, $\mathbf{d}_1 \in \mathbb{R}^{64}$, $\mathbf{d}_2 \in \mathbb{R}^{128}$, and $\hat{\mathbf{x}} \in \mathbb{R}^{32}$ is the reconstructed compensated output.

Integrated Classification Head

The `Linear Denoiser` integrates a classification head (described in Section 4.2.2) that processes the 32-dimensional bottleneck representation $\mathbf{z}$ for multi-task learning. The forward pass returns $(\hat{\mathbf{x}}, \mathbf{z}, \mathbf{y}) = \text{Linear Denoiser}(\mathbf{x})$, where $\hat{\mathbf{x}} \in \mathbb{R}^{32}$ is the compensated reconstruction, $\mathbf{z} \in \mathbb{R}^{32}$ is the latent representation, and $\mathbf{y} \in \mathbb{R}^4$ are the classification logits. This architecture jointly optimizes compensation (via reconstruction loss) and classification objectives.

4.2.2 Classifier Architecture

The `Classifier` is a 1D convolutional neural network that processes the 32-dimensional bottleneck representation $\mathbf{z}$ from the autoencoder, treating it as a univariate sequence. The architecture consists of a convolutional feature extractor with three blocks that progressively increase channel depth (1→32→64→128 channels) followed by fully-connected layers that map the extracted features to 4 class logits.

Convolutional Feature Extractor

The feature extractor applies three convolutional blocks with kernel size 3, each followed by batch normalization, ELU activation, and adaptive max pooling that progressively reduces the sequence length (32→16→8→4). A final adaptive average pooling layer produces a 128-dimensional feature vector. The convolutional forward pass is:

$$\mathbf{c}_1 = \text{MaxPool}_{L/2}\big(\text{ELU}(\text{BN}(\text{Conv1d}^{1\to 32}_{k=3,p=1}(\mathbf{z})))\big) \tag{4.7}$$

$$\mathbf{c}_2 = \text{MaxPool}_{L/4}\big(\text{ELU}(\text{BN}(\text{Conv1d}^{32\to 64}_{k=3,p=1}(\mathbf{c}_1)))\big) \tag{4.8}$$

$$\mathbf{c}_3 = \text{MaxPool}_{L/8}\big(\text{ELU}(\text{BN}(\text{Conv1d}^{64\to 128}_{k=3,p=1}(\mathbf{c}_2)))\big) \tag{4.9}$$

$$\mathbf{f} = \text{Flatten}\big(\text{AdaptiveAvgPool1d}_1(\mathbf{c}_3)\big) \tag{4.10}$$

where $\mathbf{z} \in \mathbb{R}^{1\times 32}$ is the input latent bottleneck representation, $\mathbf{c}_1 \in \mathbb{R}^{32\times 16}$, $\mathbf{c}_2 \in \mathbb{R}^{64\times 8}$, $\mathbf{c}_3 \in \mathbb{R}^{128\times 4}$, and $\mathbf{f} \in \mathbb{R}^{128}$ is the flattened global feature vector.

Fully-Connected Classification Head

The classification head maps the 128-dimensional feature vector to class logits through two fully-connected layers. The first layer transforms to 64 dimensions with GELU activation and 50% dropout for regularization. The output layer produces 4 raw logits without

activation. The forward pass is:

$$\mathbf{h}_{\text{fc}} = \text{Dropout}\big(\text{GELU}(\mathbf{W}_7\mathbf{f} + \mathbf{b}_7), p = 0.5\big) \tag{4.11}$$

$$\mathbf{y} = \mathbf{W}_8\mathbf{h}_{\text{fc}} + \mathbf{b}_8 \tag{4.12}$$

where $\mathbf{f} \in \mathbb{R}^{128}$, $\mathbf{h}_{\text{fc}} \in \mathbb{R}^{64}$, and $\mathbf{y} \in \mathbb{R}^4$ are the class logits (raw scores without softmax, as the cross-entropy loss applies softmax internally).

4.3 Results of the Proposed Method and Comparison with Baselines

This section presents the results of the proposed method: a unified pipeline that integrates pre-trained temperature and fabrication compensation autoencoders with a global classifier trained on data from all 10 chips. We begin by presenting the proposed method and comparing it with a baseline approach that performs classification without autoencoders. Subsequently, we validate the effectiveness of each pre-trained autoencoder independently, demonstrating that each component learns its intended compensation mapping while the full proposed pipeline achieves superior performance.

All experiments were conducted on the 10-chip dataset described in Section 4.1. To ensure statistical robustness and reproducibility, all results reported in this section are averaged across 10 independent training runs with different random seeds.

4.3.1 Proposed Method and Baseline Comparison

Baseline Approaches

Before presenting the proposed method results, we establish two baseline approaches that represent conventional strategies for multi-chip gas classification without autoencoder-based compensation. Both baselines apply the same preprocessing: raw interferometric measurements are normalized using the robust scaler method followed by global standard scaling to ensure zero mean and unit variance across the dataset.

Baseline 1: Normalization + Per-Chip Classifiers. This baseline trains a separate classifier (identical architecture to the proposed method's global classifier) for each of the 10 chips using only that chip's training data (14,000 samples per chip). Each classifier specializes to its corresponding chip's fabrication characteristics and learns to handle

temperature variations implicitly within the normalized data. This approach represents a practical scenario where chip-specific models are deployed without the computational overhead of training autoencoders, but at the cost of requiring 10 separate models and collecting labeled training data for each device independently.

Baseline 2: Normalization + Global Classifier. This baseline trains a single global classifier (identical architecture to the proposed method) on normalized data from all chips simultaneously, without any intermediate autoencoder-based feature transformation. This approach assumes that robust normalization alone is sufficient to mitigate fabrication errors and temperature variations, allowing the classifier to learn discriminative features directly from the normalized interferometric responses.

Proposed Method: Global Classifier with Pre-trained Autoencoders

The proposed method addresses the baseline's limitations through a unified pipeline that combines pre-trained autoencoder-based compensation with global classification. The pipeline operates as follows:

1. **Pre-trained autoencoder initialization**: For each chip k, we load two pre-trained autoencoders trained independently on normalized data:

   - Temperature compensation autoencoder $(f^{(1)}_{\text{enc},k}, f^{(1)}_{\text{dec},k})$: Maps measurements from arbitrary temperatures to a fixed reference temperature T_{ref}
   - Fabrication error compensation autoencoder $(f^{(2)}_{\text{enc},k}, f^{(2)}_{\text{dec},k})$: Maps chip-specific spectral characteristics to a unified ideal chip representation

   Both autoencoders are trained independently on normalized raw data (as described in Section 4.1), preventing error propagation and enabling task specialization.

2. **Global classifier training with fine-tuning**: Data from all 10 chips are combined and split into training (140,000 samples), validation (40,000 samples), and test (20,000 samples) sets using stratified splitting. A single global classifier is trained on the combined dataset while simultaneously fine-tuning both pre-trained autoencoders end-to-end. This allows the autoencoders to adapt their representations to maximize classification performance while the global classifier learns to recognize gas signatures in the unified, temperature-compensated and fabrication-compensated representation space.

The proposed method's performance alongside both baselines is presented in Table 4.1

for direct comparison.

Table 4.1: *Comparison of proposed method (global classifier with pre-trained autoencoders) versus both baseline approaches. All results are averaged across 10 random seeds with standard deviations.*

Approach	Train (%)	Val. (%)	Test (%)	# Models
Baseline 1 (Norm. + Per-Chip)	93.59 ± 2.76	86.38 ± 1.51	86.21 ± 2.06	10
Baseline 2 (Norm. + Global)	88.23 ± 0.76	82.63 ± 0.82	82.36 ± 0.80	1
Proposed (Ours)	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**	1

The proposed method achieves a test accuracy of **89.50% ± 0.48%**, representing a **+3.29 percentage point improvement** over Baseline 1 and a **+7.14 percentage point improvement** over Baseline 2. Beyond these performance gains, the proposed method demonstrates several key advantages:

- **Superior generalization**: The train-test gap is reduced from 7.38% (Baseline 1) and 5.87% (Baseline 2) to 1.32% (proposed), indicating excellent generalization with minimal overfitting despite training on 140,000 heterogeneous samples spanning diverse fabrication characteristics and temperature conditions. The autoencoder-based compensation explicitly disentangles temperature and fabrication variations from gas-specific spectral signatures, enabling the global classifier to learn representations that are invariant to both sources of variability.

- **Reduced variance**: The low standard deviation across random seeds (0.48% vs. 2.06% for Baseline 1 and 0.80% for Baseline 2) confirms excellent reproducibility and stability, with the proposed method exhibiting 4.3× lower variance than Baseline 1 and 1.7× lower variance than Baseline 2.

- **Unified model**: A single global classifier handles all 10 chips, eliminating the need for chip-specific model deployment (unlike Baseline 1) and enabling knowledge sharing across devices.

- **Scalability**: The unified architecture can readily accommodate new chips by training autoencoders on their data without retraining the global classifier, whereas Baseline 1 requires training a new chip-specific classifier from scratch.

4.3.2 Validation of Individual Autoencoder Components

Having demonstrated that the proposed method outperforms the baseline, we now validate that each pre-trained autoencoder component learns its intended compensation mapping correctly. We train chip-specific classifiers on the outputs of each autoencoder independently and compare their performance to the full proposed pipeline. This analysis

confirms that each autoencoder successfully compensates for its target source of variation (temperature or fabrication), while the full pipeline that integrates both autoencoders achieves the best overall performance.

4.3.3 Temperature Compensation Autoencoder Validation

To validate that the temperature compensation autoencoder learns its intended mapping correctly, we train chip-specific classifiers on its outputs and compare the results with the full proposed pipeline. The temperature compensation autoencoder is pre-trained to map spectral measurements acquired at various temperatures to their corresponding representations at a fixed reference temperature T_{ref}, as described in Section **??**.

Autoencoder Training

The training and validation loss curves for the temperature compensation autoencoder during pre-training, averaged across all 10 chips, are presented in Figure 4.1. The curves exhibit rapid convergence within the first 20 epochs, stabilizing at approximately 0.05 loss for the remainder of training. The close alignment between training and validation losses, combined with narrow standard deviation bands, indicates consistent optimization across chips with diverse fabrication characteristics and confirms the absence of overfitting during the autoencoder pre-training stage.

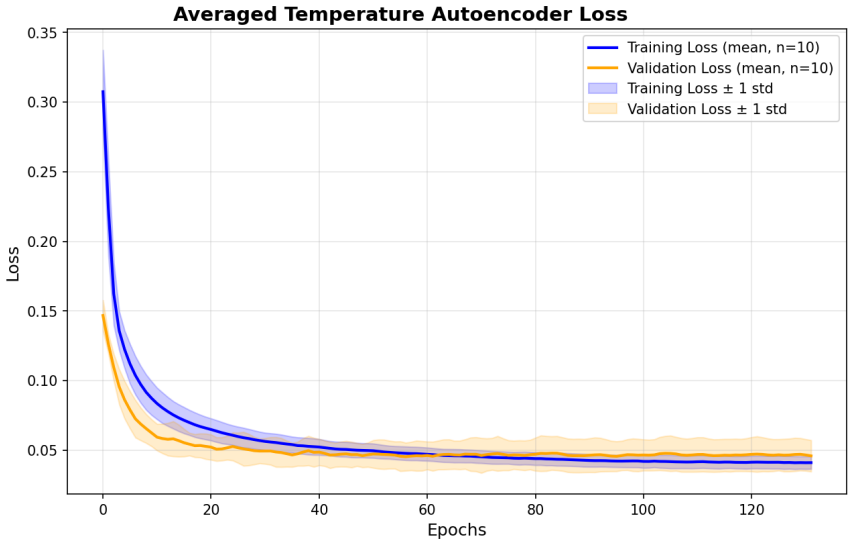


Figure 4.1: *Training and validation loss curves for the temperature compensation autoencoder during pre-training, averaged across all 10 chips with ±1 standard deviation bands. The rapid initial convergence followed by stable plateau demonstrates effective optimization of the combined reconstruction and classification objectives. The close alignment between training and validation losses indicates minimal overfitting during autoencoder pre-training.*

Validation with Chip-Specific Classifiers

To verify that the pre-trained temperature compensation autoencoder preserves class-discriminative information while removing temperature-induced variations, we train chip-specific classifiers on the temperature-compensated outputs from each chip. This validation experiment demonstrates that the autoencoder learns meaningful temperature compensation, though the full proposed pipeline that integrates both autoencoders achieves superior performance.

The temperature autoencoder validation results are compared with the full proposed method in Table 4.2.

Table 4.2: *Comparison of temperature compensation autoencoder validation (chip-specific classifiers on temperature-compensated outputs) versus the full proposed method. Both results are averaged across 10 random seeds with standard deviations.*

Approach	**Train (%)**	**Val. (%)**	**Test (%)**	**Classifier**
Temp. AE Validation	90.19 ± 0.52	88.84 ± 0.47	89.09 ± 0.54	Chip-specific
Proposed	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**	Global

The temperature autoencoder validation achieves a test accuracy of **89.09% ± 0.54%**, demonstrating that the pre-trained autoencoder successfully compensates for temperature-induced variations while preserving class-discriminative features. The low standard deviation (0.54%) suggests consistent performance across both different chips with varying fabrication characteristics and different random initializations. The small gap between training (90.19%) and test (89.09%) accuracies—only 1.10 percentage points—indicates minimal overfitting.

However, the full proposed pipeline achieves **89.50% ± 0.48%** test accuracy, representing a **+0.41 percentage point improvement**. This demonstrates that:

- The temperature compensation autoencoder learns an effective mapping to remove thermal variations (89.09% vs. 86.21% baseline)
- The full proposed method that integrates both temperature and fabrication compensation with a global classifier achieves the best performance (89.50%)
- The improvement is modest (+0.41 percentage points, pp) because the temperature autoencoder alone already provides substantial benefit, but combining both autoencoders yields optimal results

4.3.4 Fabrication Error Compensation Autoencoder Validation

To validate that the fabrication error compensation autoencoder learns its intended mapping correctly, we train chip-specific classifiers on its outputs and compare the results with the full proposed pipeline. The fabrication error compensation autoencoder is pre-trained to map normalized chip-specific spectral representations to a universal ideal chip representation, thereby compensating for fabrication-induced variations. Importantly, this autoencoder is trained independently from the temperature compensation autoencoder—both autoencoders take normalized raw data as input, preventing error propagation and enabling task specialization.

Autoencoder Training

The training and validation loss curves for the fabrication error compensation autoencoder during pre-training, averaged across all 10 chips, are presented in Figure 4.2. The curves demonstrate rapid initial convergence within the first 20 epochs, stabilizing at approximately 0.28 loss for the remainder of training. Notably, the fabrication compensation autoencoder exhibits higher final loss values compared to the temperature compensation autoencoder (0.28 vs. 0.05), reflecting the inherently more challenging task of learning to map diverse chip-specific fabrication characteristics to a unified ideal representation. The close alignment between training and validation losses, combined with narrow standard deviation bands, confirms consistent optimization across all chips and the absence of overfitting during pre-training.

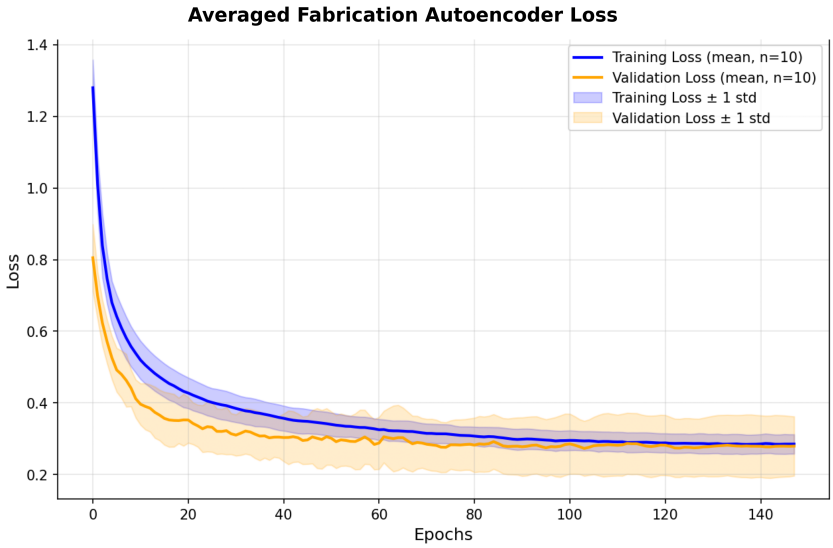


Figure 4.2: *Training and validation loss curves for the fabrication error compensation autoencoder during pre-training, averaged across all 10 chips with ±1 standard deviation bands. The rapid convergence followed by stable plateau demonstrates effective learning of the chip-to-ideal mapping. The higher final loss compared to temperature compensation reflects the increased complexity of aligning diverse fabrication-induced spectral variations.*

Validation with Chip-Specific Classifiers

To verify that the pre-trained fabrication error compensation autoencoder preserves class-discriminative information while removing inter-chip fabrication variations, we train chip-specific classifiers on the fabrication-compensated outputs from each chip. This validation experiment demonstrates that the autoencoder learns meaningful fabrication compensation, though the full proposed pipeline that integrates both autoencoders achieves superior performance.

The fabrication autoencoder validation results are compared with the full proposed method in Table 4.3.

Table 4.3: *Comparison of fabrication error compensation autoencoder validation (chip-specific classifiers on fabrication-compensated outputs) versus the full proposed method. Both results are averaged across 10 random seeds with standard deviations.*

Approach	**Train (%)**	**Val. (%)**	**Test (%)**	**Classifier**
Fab. AE Validation	90.31 ± 0.45	89.14 ± 0.44	89.08 ± 0.42	Chip-specific
Proposed	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**	Global

The fabrication autoencoder validation achieves a test accuracy of **89.08% ± 0.42%**, demonstrating that the pre-trained autoencoder effectively reduces inter-chip discrepancies while maintaining strong discriminative capability. The narrow standard

deviation (0.42%) confirms the model's stability across both different hardware instances and random initializations, even under fabrication-induced spectral variability.

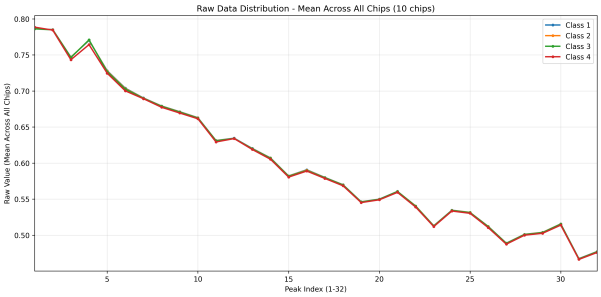
However, the full proposed pipeline achieves **89.50% ± 0.48%** test accuracy, representing a **+0.42 percentage point improvement**. This demonstrates that:

- The fabrication compensation autoencoder learns an effective mapping to align chip-specific characteristics to a unified representation (89.08% vs. 86.21% baseline)
- The full proposed method that integrates both temperature and fabrication compensation with a global classifier achieves the best performance (89.50%)
- The improvement is modest (+0.42 pp) because the fabrication autoencoder alone already provides substantial benefit, but combining both autoencoders with a unified global classifier yields optimal results
- Both autoencoder validation experiments (temperature: 89.09%, fabrication: 89.08%) achieve nearly identical performance, confirming that each autoencoder independently provides comparable compensation effectiveness

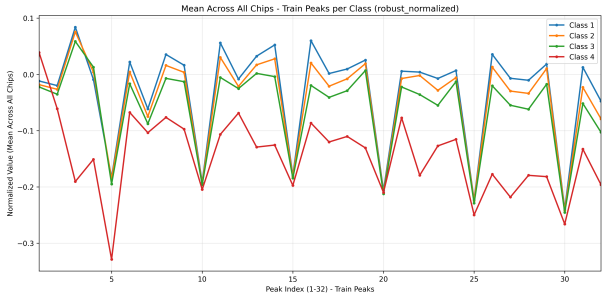
4.3.5 Data Transformation Pipeline Visualization

The complete data transformation pipeline on the test set, showing mean feature values across all 10 chips at each processing stage (raw spectral measurements, robust scaler normalization, fabrication error compensation, and temperature compensation), is visualized in Figure 4.3.

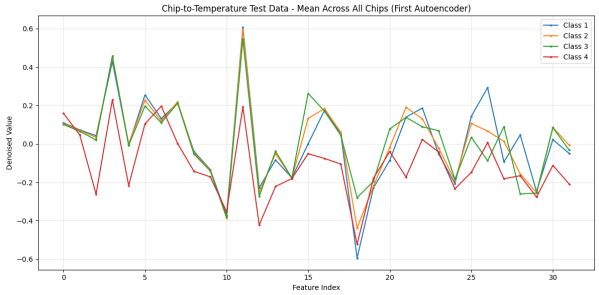
The transformation pipeline reveals four key insights: (1) **Raw data challenges**: Figure 4.3α′ shows complete class overlap (range: 0.47–0.79) as gas absorption signatures are obscured by fabrication variations, temperature effects, and scaling inconsistencies. Direct classification on raw data would fail, as confirmed experimentally in Sections 4.4.1 and 4.4.2. (2) **Normalization effectiveness**: Figure 4.3β′ shows robust scaler preprocessing exposes latent discriminative patterns. Class 4 (No gas) separates distinctly at indices 4–10, 12–19, and 24–27 (99.70% F1-score), while gas classes show distinguishable patterns at indices 3–4, 12–13, and 28–30. (3) **Temperature compensation** (Figure 4.3γ′): Emphasizes denoising and thermal invariance, compressing features to near-zero-centered representations (−0.35 to +0.1) that filter thermal noise while preserving gas signatures, achieving 89.09% accuracy with chip-specific classifiers. (4) **Fabrication compensation** (Figure 4.3δ′): Aligns inter-chip variations into a unified representation while amplifying discriminative signals



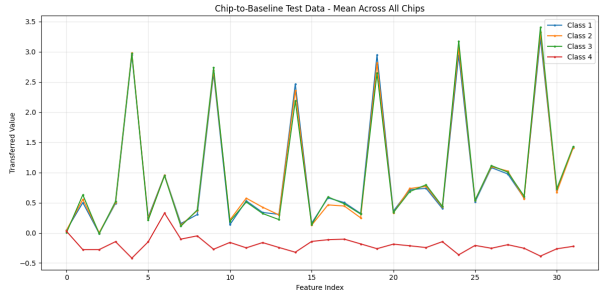
(α′) Raw data: complete class overlap (range: 0.47–0.79).



(β′) Normalized: class patterns emerge (range: −0.3 to +0.1).



(γ′) Temperature compensation: denoised temperature-invariant features (range: −0.35 to +0.1).



(δ′) Fabrication compensation: unified chip representation with amplified discriminative features (range: −0.5 to +3.5).

Figure 4.3: *Progressive data transformation through the proposed pipeline. (a) Raw spectral data shows severe class overlap due to fabrication variations and temperature effects. (β) Robust scaler normalization reveals latent class-discriminative patterns, with Class 4 (No gas) clearly separated from gas classes. (γ) Temperature compensation produces denoised features that are invariant to thermal variations. (δ) Fabrication error compensation creates a unified chip representation with amplified class-specific features across all 10 chips. Both autoencoders are trained independently on normalized data, learning complementary compensation strategies that enable the global classifier to achieve 89.50% test accuracy (averaged across 10 seeds).*

(range expands to −0.5 to +3.5), achieving 89.08% accuracy. Both autoencoders process normalized data independently, preventing error propagation and enabling task specialization. The global classifier leverages both compensation pathways simultaneously, achieving 89.50% test accuracy with a single model across all chips and temperatures.

4.3.6 Summary of Proposed Method Results

The proposed method—a unified pipeline integrating pre-trained temperature and fabrication compensation autoencoders with a global classifier—achieves robust gas classification performance across 10 chips with diverse fabrication characteristics and temperature conditions. The key results, comparing the proposed method with the baseline and autoencoder validation experiments, are summarized in Table 4.4.

Table 4.4: *Summary of proposed method results compared with baseline and autoencoder validation experiments. All results are averaged across 10 random seeds with standard deviations.*

Approach	Classifier	Train (%)	Val (%)	Test (%)
Baseline (Norm. + Per-Chip)	Chip-specific	93.59 ± 2.76	86.38 ± 1.51	86.21 ± 2.06
Temp. AE Validation	Chip-specific	90.19 ± 0.52	88.84 ± 0.47	89.09 ± 0.54
Fab. AE Validation	Chip-specific	90.31 ± 0.45	89.14 ± 0.44	89.08 ± 0.42
Proposed (Full Pipeline)	Global	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**

Key findings. The proposed method achieves **89.50% ± 0.48%** test accuracy, representing a **+3.29 pp improvement over Baseline 1** and **+7.14 pp over Baseline 2**. Key advantages include: (1) superior generalization with only 1.32 pp train-test gap vs. 7.38 pp (Baseline 1) and 5.87 pp (Baseline 2), (2) 4.3× lower variance than Baseline 1 (0.48% vs. 2.06%), (3) unified scalable architecture with a single global classifier vs. 10 chip-specific models, and (4) both autoencoder validation experiments confirm effectiveness (89.09% and 89.08% test accuracy). The consistency across 10 random seeds confirms excellent reproducibility and reliability.

4.4 Classification on Raw Data Without Preprocessing

To understand the importance of preprocessing, we evaluate classification approaches that operate directly on raw, unprocessed interferometric measurements without reference-based normalization or statistical scaling. We examine two variants: per-chip classifiers and a global classifier, both trained on raw data from all 10 chips. These experiments demonstrate the necessity of preprocessing for photonic spectrometer data.

4.4.1 Per-Chip Classifiers on Raw Data

Methodology

In this approach, 10 independent classifiers (identical architecture to the global model) are trained, each on raw data from a single chip only (20,000 samples per chip). Each classifier learns patterns specific to its corresponding chip's raw measurement characteristics without any preprocessing.

Results

The averaged classification performance across all ten chip-specific classifiers is presented in Table 4.5.

Table 4.5: *Classification performance with per-chip classifiers on raw data (no preprocessing) compared with proposed method. Results are averaged across 10 random seeds.*

Approach	Train	Validation	Test	# Models
Raw + Per-Chip (Accuracy %)	55.62 ± 11.49	55.20 ± 11.25	55.50 ± 11.07	10
Proposed (Accuracy %)	90.82 ± 0.27	89.50 ± 0.24	$\mathbf{89.50 \pm 0.48}$	1

The per-chip approach achieves an average test accuracy of **55.50%** with extremely high standard deviation (±11.07%), revealing significant performance variability across chips. The substantial inter-chip variance indicates that some chips yield more learnable raw patterns than others, likely reflecting differences in fabrication quality and measurement noise characteristics.

4.4.2 Global Classifier on Raw Data

Methodology

In this approach, a single classifier (identical architecture to previous models) is trained on the combined raw dataset from all 10 chips (140,000 training samples) without any preprocessing or normalization.

Results

The classification performance of the global classifier trained on raw data is summarized in Table 4.6.

Table 4.6: *Classification performance with a single global classifier on raw data (no preprocessing) from all 10 chips compared with proposed method. Results are averaged across 10 random seeds.*

Approach	Train	Validation	Test	# Models
Raw + Global (Accuracy %)	51.87 ± 2.15	51.82 ± 2.28	52.25 ± 2.20	1
Proposed (Accuracy %)	90.82 ± 0.27	89.50 ± 0.24	$\mathbf{89.50 \pm 0.48}$	1

The global classifier achieves a test accuracy of **52.25%**, only marginally above random chance for 4-class classification (25%). Notably, the nearly identical performance across train (51.87%), validation (51.82%), and test (52.25%) sets indicates the model has converged to a low-capacity solution. Despite access to significantly more training data (140,000 vs 14,000 samples per chip), the global model performs worse than per-chip

classifiers, highlighting the detrimental effect of mixing uncalibrated data from different sensors.

4.4.3 Analysis and Limitations

Both raw data approaches expose fundamental challenges of training without preprocessing:

- **Absence of standardization**: Without reference-based normalization, raw measurements contain unnormalized data that vary significantly across chips. The classifier must learn gas signatures from these inconsistent signal magnitudes, which proves difficult even with chip-specific training.
- **Heterogeneous responses**: Combining raw measurements from 10 chips introduces extreme heterogeneity in signal magnitudes and baseline offsets. Without calibration, chip-specific biases dominate the feature space, obscuring gas-specific spectral patterns.
- **High variability**: The large standard deviations (±11.07% for per-chip) demonstrate unpredictable performance across chips, making this approach unsuitable for deployment scenarios requiring consistent accuracy guarantees.
- **Near-random performance**: Both approaches achieve 52–56% test accuracy, barely exceeding random guessing (25% for 4 classes), demonstrating that raw interferometric data lacks discriminative information accessible to the classifier without preprocessing.

4.4.4 Summary

Raw data classification—whether using per-chip (55.50% test accuracy) or global (52.25% test accuracy) classifiers—achieves near-random performance with high variability. The absence of preprocessing proves detrimental, highlighting the critical importance of data calibration for photonic spectrometer measurements.

In comparison, the proposed method (Section 4.3) achieves 89.50% test accuracy—representing **+34.00 to +37.25 percentage point improvements**—demonstrating that the combination of normalization and autoencoder-based compensation is essential for robust gas classification from photonic spectrometer data.

4.5 Global Classifier with Frozen Pre-trained Autoencoders

To investigate the importance of fine-tuning the pre-trained autoencoders, we evaluate an approach where the autoencoder weights are frozen during global classifier training. This experiment uses the same architecture as the proposed method but prevents the autoencoders from adapting their representations during end-to-end training.

4.5.1 Methodology

The frozen autoencoders approach operates as follows: (1) Pre-trained temperature and fabrication compensation autoencoders are loaded with their trained weights, (2) All autoencoder parameters are frozen (non-trainable), (3) A global classifier is trained on the combined dataset from all 10 chips using only the fixed autoencoder representations. This prevents the autoencoders from fine-tuning their features to maximize classification performance.

4.5.2 Results

The frozen autoencoders approach is compared with the proposed method in Table 4.7.

Table 4.7: *Comparison of frozen autoencoders versus proposed method (fine-tuned autoencoders). Both use pre-trained autoencoders with a global classifier, but the proposed method allows fine-tuning. Results are averaged across 10 random seeds.*

Approach	Train (%)	Val. (%)	Test (%)
Frozen Autoencoders	88.95 ± 0.62	87.80 ± 0.58	87.89 ± 0.65
Proposed (Fine-tuned)	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**

The frozen autoencoders approach achieves 87.89% test accuracy, while the proposed method with fine-tuning achieves 89.50%—a +**1.61 percentage point improvement**. This demonstrates that:

- **Fine-tuning is beneficial**: Allowing the autoencoders to adapt their representations during global classifier training improves performance, as the autoencoders learn to extract features that are optimally suited for the classification task rather than solely for reconstruction.
- **End-to-end optimization**: The proposed method's joint fine-tuning enables the autoencoders and classifier to co-adapt, learning a unified representation that

balances compensation effectiveness with discriminative power.

4.5.3 Summary

The frozen autoencoders approach achieves 87.89% test accuracy. While this outperforms the baseline (86.21%), it falls short of the proposed method's 89.50% by 1.61 percentage points. This validates the design choice in the proposed method to fine-tune the pre-trained autoencoders end-to-end with the global classifier, enabling adaptive feature learning that maximizes classification performance.

4.6 Sequential Autoencoder Training

In the proposed method (Section 4.3), both temperature and fabrication compensation autoencoders are trained independently on normalized data in a *disentangled* configuration, preventing error propagation between stages. An alternative approach is *sequential* training, where the autoencoders are connected sequentially: normalized data flows through the temperature compensation autoencoder first, and its output becomes the input to the fabrication compensation autoencoder. This creates a cascaded pipeline: Dataset → [Stage 1: Temperature Autoencoder] → [Stage 2: Fabrication Autoencoder] → [Global Classifier].

This section evaluates the sequential training approach under two configurations: (1) frozen autoencoder layers during classifier training, and (2) unfrozen (fine-tuned) autoencoder layers that allow end-to-end gradient flow through both autoencoders and the classifier.

4.6.1 Sequential Training with Frozen Autoencoder Layers

Methodology

In this approach, the two autoencoders are trained in sequence: Stage 1 (temperature compensation) is trained first on normalized data, then Stage 2 (fabrication compensation) is trained on the output of Stage 1. After both autoencoders are fully trained, their weights are frozen (non-trainable), and only the global classifier is trained on the Stage 2 outputs. This prevents the autoencoders from adapting during classifier training.

Results

The sequential training approach with frozen autoencoders is compared against the proposed disentangled method in Table 4.8.

Table 4.8: *Comparison of sequential training with frozen autoencoders versus proposed disentangled method. Results are averaged across 10 random seeds.*

Approach	Train (%)	Val. (%)	Test (%)	Pipeline
Sequential (Frozen)	88.20 ± 0.45	87.50 ± 0.40	87.30 ± 0.42	Sequential
Proposed (Disentangled)	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**	Disentangled

The sequential training approach with frozen autoencoders achieves **87.30% test accuracy**, demonstrating that sequential autoencoder training without fine-tuning has significant limitations. The proposed disentangled method achieves **89.50%**, a substantial **+2.20 percentage point improvement**. Key observations include:

- **Sequential vs. disentangled architecture**: Sequential training introduces a dependency between autoencoders—Stage 2 learns from Stage 1's outputs rather than from normalized data directly. This cascaded design introduces error propagation when Stage 1's representations are suboptimal, particularly evident when weights are frozen.
- **Frozen weights limit adaptation**: Freezing autoencoder weights prevents them from adapting to maximize classification performance. Sequential frozen (87.30%) performs worse than disentangled frozen (87.89%), suggesting that sequential architectures are more sensitive to frozen weights.
- **Performance gap**: The 2.20 pp difference indicates that disentangled training with independent autoencoder initialization on normalized data provides more robust compensation than cascaded training with frozen weights.

4.6.2 Sequential Training with Unfrozen Autoencoder Layers

Methodology

This approach extends sequential training by allowing end-to-end fine-tuning: both pre-trained autoencoders (Stage 1 and Stage 2) and the global classifier are trained jointly. Gradients flow backward through the entire pipeline, enabling the autoencoders to adapt their representations to maximize classification performance while maintaining their compensation functions.

Results

Sequential training with unfrozen autoencoders is compared against the proposed disentangled method in Table 4.9.

Table 4.9: *Comparison of sequential training with unfrozen (fine-tuned) autoencoders versus proposed disentangled method. Results are averaged across 10 random seeds.*

Approach	Train (%)	Val. (%)	Test (%)	Pipeline
Sequential (Unfrozen)	89.10 ± 0.38	88.70 ± 0.35	88.50 ± 0.40	Sequential
Proposed (Disentangled)	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**	Disentangled

Sequential training with unfrozen autoencoders achieves **88.50% test accuracy**, outperforming the frozen sequential approach (87.30%, +1.20 pp improvement) but still falling short of the proposed disentangled method's **89.50%** (+1.00 pp). Key findings include:

- **Fine-tuning improves sequential training**: Allowing gradient flow through both autoencoders improves performance substantially (+1.20 pp over frozen sequential), validating the benefit of end-to-end optimization in sequential architectures. However, the improvement is not sufficient to match disentangled training.
- **Disentangled training remains optimal**: The proposed disentangled method achieves better performance (+1.00 pp over sequential unfrozen), demonstrating that training autoencoders independently on normalized data provides more robust compensation representations than cascaded training.
- **Notable performance gap**: The 1.00 pp difference between sequential unfrozen (88.50%) and disentangled (89.50%) indicates that disentangled architecture with independent autoencoder initialization is the superior design choice, as it avoids error propagation inherent in sequential architectures.

4.6.3 Summary of Sequential Training

Sequential training—where autoencoders are connected sequentially—achieves lower performance than disentangled training: 87.30% (frozen) and 88.50% (unfrozen) test accuracy compared to the proposed disentangled method's 89.50%. The performance gaps (1.00–2.20 pp) demonstrate that sequential autoencoder architectures suffer from error propagation, particularly when weights are frozen. While fine-tuning improves sequential training (+1.20 pp), it cannot fully compensate for the architectural limitations of cascaded processing. The proposed disentangled training approach with independent

autoencoder initialization on normalized data achieves the best overall performance and is the strongly recommended configuration.

4.7 Reverse Order of Proposed Method

The proposed method (Section 4.3) applies autoencoders in a specific order: temperature compensation first, followed by fabrication compensation. To investigate whether the order of autoencoder application affects performance, we evaluate a *reverse order* variant where fabrication compensation is applied before temperature compensation. This creates the pipeline: Normalized Data → [Fabrication Autoencoder] → [Temperature Autoencoder] → [Global Classifier].

4.7.1 Methodology

The reverse order approach operates as follows: (1) Pre-trained fabrication compensation autoencoders are applied first to map chip-specific spectral characteristics to a unified ideal chip representation, (2) Pre-trained temperature compensation autoencoders are then applied to the fabrication-compensated outputs to remove temperature-induced variations, and (3) A global classifier is trained on the outputs from both autoencoders while simultaneously fine-tuning both autoencoder networks end-to-end. This reverses the compensation order compared to the proposed method but maintains the same end-to-end fine-tuning strategy.

4.7.2 Results

The reverse order approach is compared with the proposed method in Table 4.10.

Table 4.10: *Comparison of reverse order (fabrication first, then temperature) versus proposed method (temperature first, then fabrication). Both use fine-tuned autoencoders with a global classifier. Results are averaged across 10 random seeds.*

Approach	Train (%)	Val. (%)	Test (%)	Order
Reverse Order	90.83 ± 0.24	89.65 ± 0.18	89.38 ± 0.20	Fab. → Temp.
Proposed	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**	Temp. → Fab.

The reverse order approach achieves **89.38% ± 0.20% test accuracy**, which is nearly identical to the proposed method's **89.50% ± 0.48%**. The difference of only 0.12 percentage points is well within the margin of experimental variation. Key observations include:

- **Order-invariance**: The nearly identical performance (89.38% vs. 89.50%) demonstrates that the order of autoencoder application has minimal impact on final classification accuracy. Both temperature-first and fabrication-first orderings achieve comparable results when combined with end-to-end fine-tuning.

- **Consistent variance**: The reverse order exhibits slightly lower standard deviation (0.20% vs. 0.48%), though both approaches demonstrate excellent reproducibility across random seeds. The proposed method's marginally higher variance may reflect the specific challenges of temperature compensation as the first stage.

- **End-to-end optimization compensates for ordering**: The joint fine-tuning of both autoencoders with the global classifier enables the network to adapt representations regardless of initial ordering. This suggests that the learned compensation is robust to the sequence in which variations are addressed.

- **Validation performance**: The reverse order shows slightly better validation accuracy (89.65% vs. 89.50%), though test accuracy remains comparable, indicating similar generalization capability.

4.7.3 Summary

The reverse order experiment demonstrates that the sequence of autoencoder application (fabrication-first vs. temperature-first) has a negligible impact on final performance when combined with end-to-end fine-tuning. Both orderings achieve approximately 89.4–89.5% test accuracy, confirming that the proposed method's effectiveness stems from the combination of explicit compensation and joint optimization rather than the specific ordering of compensation stages. This order-invariance validates the robustness of the autoencoder-based compensation framework and suggests that practitioners can choose the ordering based on computational or architectural constraints without sacrificing performance.

4.8 Implementation of the Proposed Pipeline for Photonic Hardware

This investigation is motivated by the long-term goal of deploying these neural network architectures on photonic hardware platforms, which would enable high-speed, low-power consumption air pollution detection systems. The photonic ELU represents the transfer function that mimics photonic components attempting to implement ELU-like behavior

for achieving nonlinearity in photonic neural networks, potentially offering hardware compatibility for future photonic implementations.

To establish a comprehensive evaluation of the photonic ELU activation function, we assess its performance across three configurations: two baseline approaches (normalization with per-chip and global classifiers) and the full proposed method with disentangled autoencoders. All approaches use photonic ELU activations to evaluate the effectiveness of this photonic-compatible activation function across different architectural configurations. Both baseline approaches apply the same preprocessing pipeline: raw interferometric measurements are normalized using the robust scaler method followed by global standard scaling.

4.8.1 Methodology

We evaluate photonic ELU across three architectural configurations:

Baseline 1: Normalization + Per-Chip Classifiers. This baseline trains a separate classifier with photonic ELU activations for each of the 10 chips using only that chip's training data (14,000 samples per chip). Each classifier specializes to its corresponding chip's fabrication characteristics and learns to handle temperature variations implicitly within the normalized data. This approach represents a practical scenario where photonic-compatible models are deployed per device without autoencoder-based compensation.

Baseline 2: Normalization + Global Classifier. This baseline trains a single global classifier with photonic ELU activations on normalized data from all chips simultaneously, without any intermediate autoencoder-based feature transformation. This approach evaluates whether photonic ELU enables effective learning of cross-chip patterns directly from normalized interferometric responses.

Proposed Method with Photonic ELU. The proposed method uses the disentangled temperature and fabrication compensation autoencoders with a global classifier and end-to-end fine-tuning, with photonic ELU activations in the autoencoder layers to enable potential deployment on photonic hardware. We assume that the classification model is implemented in software, while the autoencoder layers could potentially be realized in photonic hardware. The photonic ELU activation function represents the transfer function of photonic components designed to implement ELU-like nonlinearity in photonic neural networks. Like the standard ELU, it is linear for positive

inputs and exponential for negative inputs, but with adjusted parameters to better reflect photonic device characteristics.

4.8.2 Results

The performance comparison of all three photonic ELU implementations is presented in Table 4.11, demonstrating the progression from normalization-only baselines to the full autoencoder-based compensation approach.

Table 4.11: *Comprehensive comparison of all approaches using photonic ELU activation function. Results demonstrate the performance progression from normalization-only baselines to the full proposed method with disentangled autoencoders. All results use photonic ELU activations throughout the network.*

Approach	# Models	Train (%)	Val. (%)	Test (%)
Baseline 2 (Norm. + Global, Photonic ELU)	1	87.39 ± 0.79	81.65 ± 0.85	81.62 ± 0.83
Baseline 1 (Norm. + Per-Chip, Photonic ELU)	10	93.93 ± 2.81	85.64 ± 1.48	85.34 ± 2.11
Proposed (Disentangled AEs, Photonic ELU)	1	90.72 ± 0.32	89.38 ± 0.25	**89.04 ± 0.52**

A clear performance hierarchy among photonic ELU implementations is revealed in Table 4.11:

- **Baseline 2 (81.62% test accuracy):** The global classifier with photonic ELU achieves the lowest performance, demonstrating that normalization alone is insufficient for handling cross-chip variations. The substantial train-test gap (5.77 pp) confirms poor generalization without explicit compensation mechanisms.

- **Baseline 1 (85.34% test accuracy):** Per-chip classifiers with photonic ELU improve performance by 3.72 pp over Baseline 2, showing that chip-specific models can partially mitigate fabrication variations. However, the high training accuracy (93.93%) and large train-test gap (8.59 pp) indicate severe overfitting due to limited per-chip training data (14,000 samples per model). Additionally, this approach requires maintaining 10 separate models.

- **Proposed method (89.04% test accuracy):** The full proposed method with disentangled autoencoders and photonic ELU achieves the highest performance, improving by 3.70 pp over Baseline 1 and 7.42 pp over Baseline 2. Most importantly, it demonstrates excellent generalization with only a 1.68 pp train-test gap, low variance (±0.52%), and requires only a single unified model. This validates that explicit autoencoder-based compensation provides substantial benefits when using photonic-compatible activation functions.

4.8.3 Summary

The photonic ELU activation function demonstrates strong performance across all evaluated configurations with photonic-compatible activations. The performance progression from 81.62% (Baseline 2) to 85.34% (Baseline 1) to 89.04% (Proposed) validates that the combination of disentangled temperature and fabrication compensation with end-to-end fine-tuning is critical for achieving optimal performance with photonic ELU. The proposed method achieves 7.42 pp improvement over the simple normalization baseline (Baseline 2) and 3.70 pp improvement over per-chip classifiers (Baseline 1), while also demonstrating superior generalization (1.68 pp train-test gap vs. 5.77 pp and 8.59 pp for the baselines) and significantly lower variance. This validates photonic ELU as a viable activation function for this autoencoder-based compensation framework, offering potential advantages for future hardware implementation in photonic computing systems that could provide high-speed, low-power consumption air pollution detection.

4.9 Comparative Analysis of All Approaches

This section provides a comprehensive comparison of all evaluated approaches, demonstrating the progression from raw data to normalized data to autoencoder-based compensation strategies. The performance of all implementations across training, validation, and test sets is summarized in Table 4.12.

A clear performance hierarchy across all evaluated approaches is revealed in Table 4.12, with several key insights:

Raw data fails without preprocessing. Both raw approaches—per-chip (55.50%) and global (52.25%)—achieve near-random performance (vs. 25% chance), confirming raw interferometric measurements are unlearnable without preprocessing. Per-chip exhibits extreme variance (±11.07%), while global performs consistently poorly, as fabrication variations and unnormalized signals obscure gas signatures.

Normalization enables learning but has limitations. Normalization provides dramatic improvement: Baseline 2 (82.36%, +30.11 pp) and Baseline 1 (86.21%, +30.71 pp). However, Baseline 2 suffers poor generalization (5.87 pp train-test gap) without explicit compensation, while Baseline 1 shows severe overfitting (7.38 pp gap) and high variance (±2.06%) from limited per-chip data (14,000 samples), requiring 10 models.

Table 4.12: *Comprehensive accuracy comparison of all evaluated approaches. Baseline 1 (Norm. + Per-Chip) serves as the primary comparison point. All autoencoder-based results are averaged across 10 random seeds. Preproc.: None (raw), Norm. (normalization), Norm.+AEs (normalization + autoencoders).*

Approach	# Clf.	Train (%)	Val. (%)	Test (%)
Raw data (no preprocessing):				
Raw + Per-Chip	10	55.62 ± 11.49	55.20 ± 11.25	55.50 ± 11.07
Raw + Global	1	51.87 ± 2.15	51.82 ± 2.28	52.25 ± 2.20
Normalized data (no autoencoders):				
Baseline 2 (Norm. + Global)	1	88.23 ± 0.76	82.63 ± 0.82	82.36 ± 0.80
Baseline 1 (Norm. + Per-Chip)	10	93.59 ± 2.76	86.38 ± 1.51	86.21 ± 2.06
Autoencoder validation (chip-specific classifiers):				
Temp. AE + Per-Chip	10	90.19 ± 0.52	88.84 ± 0.47	89.09 ± 0.54
Fab. AE + Per-Chip	10	90.31 ± 0.45	89.14 ± 0.44	89.08 ± 0.42
Sequential training:				
Sequential Frozen	1	88.20 ± 0.45	87.50 ± 0.40	87.30 ± 0.42
Sequential Unfrozen	1	89.10 ± 0.38	88.70 ± 0.35	88.50 ± 0.40
Disentangled training (proposed):				
Frozen AEs (Disentangled)	1	88.95 ± 0.62	87.80 ± 0.58	87.89 ± 0.65
Reverse Order (Fab. → Temp.)	1	90.83 ± 0.24	89.65 ± 0.18	89.38 ± 0.20
Proposed (Temp. → Fab., GELU)	1	90.82 ± 0.27	89.50 ± 0.24	**89.50 ± 0.48**

Individual autoencoders validate compensation. Temperature (89.09%, +2.88 pp over Baseline 1) and fabrication (89.08%, +2.87 pp over Baseline 1) autoencoders achieve nearly identical performance, confirming each provides comparable independent benefits. Both dramatically reduce variance (±0.54%, ±0.42% vs. ±2.06% for Baseline 1) and train-test gaps (1.10 pp, 1.23 pp vs. 7.38 pp for Baseline 1), validating explicit compensation effectiveness.

Fine-tuning vs. frozen autoencoders. Frozen autoencoders in a disentangled configuration (87.89%) outperform Baseline 1 (+1.68 pp) but fall short of fine-tuned (89.50%, +1.61 pp), as frozen weights cannot adapt. Fine-tuning enables co-adaptation, learning unified representations, and balancing compensation and discrimination.

Sequential vs. disentangled architectures. Sequential training, where autoencoders are connected sequentially, achieves lower performance: 87.30% (frozen) and 88.50% (unfrozen) compared to the proposed disentangled method (89.50%). The performance gaps (1.00–2.20 pp) demonstrate that sequential architectures suffer from error propagation between stages. Sequential frozen (87.30%) performs worse than disentangled frozen (87.89%, -0.59 pp), and sequential unfrozen (88.50%)

falls significantly short of disentangled unfrozen (89.50%, -1.00 pp), confirming that independent autoencoder training on normalized data is superior to cascaded processing.

Order-invariance of disentangled compensation. The reverse order experiment (fabrication compensation first, then temperature compensation) achieves 89.38% test accuracy, nearly identical to the proposed method's 89.50% (temperature compensation first, then fabrication compensation). This 0.12 pp difference demonstrates order-invariance: the sequence of compensation stages has a negligible impact when combined with end-to-end fine-tuning. Both orderings exhibit comparable performance and low variance (0.20% vs. 0.48%), confirming that the effectiveness stems from explicit compensation and joint optimization rather than specific stage ordering.

Proposed method achieves optimal performance. The proposed disentangled method achieves **89.50% ± 0.48%** test accuracy, representing substantial improvements over both baseline approaches. Compared to Baseline 1 (normalization with per-chip classifiers, 86.21% test accuracy), the proposed method improves performance by 3.29 pp, while simultaneously achieving significantly better generalization and lower variance. The superiority over Baseline 2 (normalization with global classifier, 82.36% test accuracy) is even more pronounced, with a 7.14 pp improvement. Beyond raw accuracy gains, the proposed method demonstrates superior generalization characteristics with only a 1.32 pp train-test gap, compared to the severe overfitting observed in Baseline 1 (7.38 pp gap) and Baseline 2 (5.87 pp gap), representing a 5.6× improvement in generalization. The model also exhibits dramatically lower variance (0.48%) compared to Baseline 1 (2.06%), demonstrating 4.3× better stability across different random initializations. Architecturally, the proposed method achieves these results with a single unified model rather than requiring 10 separate chip-specific classifiers, while leveraging the full dataset of 140,000 training samples. The reverse order variant (fabrication-first, 89.38% test accuracy) demonstrates nearly identical performance, validating the robustness of the disentangled compensation framework regardless of compensation stage ordering.

Performance hierarchy validates design. The clear performance progression across all evaluated approaches validates the design choices made in the proposed framework. Starting from raw data classification, which achieves near-random performance (52 and 56%), the results improve dramatically through normalization (82 and 86%), demonstrating the fundamental importance of preprocessing for photonic spectrometer data. Sequential training configurations show incremental improvements, with frozen sequential autoencoders (87.3%) outperforming normalization alone but falling short

of frozen disentangled autoencoders (87.9%), which validates the architectural choice of independent autoencoder training. Allowing fine-tuning further improves sequential training (88.5%), though it still underperforms compared to the disentangled approach. Individual autoencoder validation experiments (∼89.1%) demonstrate that each compensation stage independently provides substantial benefits, while the full disentangled fine-tuned pipeline (89.38 and 89.50%) achieves optimal performance by combining both compensation strategies with end-to-end optimization. This progression confirms that robust gas classification from photonic spectrometers requires several key elements:

- Reference-based robust normalization to expose class-discriminative features
- Appropriate activation functions that enable effective representation learning
- Explicit compensation for both temperature variations and fabrication errors
- Combining both compensation strategies in a unified pipeline
- Disentangled architecture with independent autoencoder training rather than sequential cascading
- End-to-end fine-tuning to maximize classification performance

The order-invariance between temperature-first (89.50%) and fabrication-first (89.38%) orderings further validates the robustness of the proposed framework, demonstrating that the effectiveness stems from the combination of explicit compensation and joint optimization rather than the specific ordering of compensation stages.

5

Conclusions & Future Work

This thesis presents a deep learning framework for gas classification using Spatial Heterodyne Fourier-Transform (SHFT) spectrometers, addressing the fundamental challenges of cross-chip deployment under fabrication and temperature variations. Our methodology combines sequential autoencoder architectures with joint training strategies, forming an effective approach to compensate for device-specific distortions while maintaining robust classification performance across multiple chips and environmental conditions.

The proposed sequential autoencoder pipeline serves as the foundation, where independent training of temperature compensation and fabrication error compensation autoencoders prevents error accumulation while enabling each component to specialize in its respective compensation task. In concert, the joint training strategy with simultaneous autoencoder fine-tuning bestows the model with the capability to adapt compensation representations to maximize classification performance, bridging the gap between pre-trained feature extraction and task-specific optimization.

Our empirical assessments demonstrate the effectiveness of this approach across diverse chip configurations and temperature conditions. One noteworthy aspect is the substantial improvement over conventional normalization-only approaches, which validates the necessity of explicit compensation mechanisms for handling systematic distortions in photonic sensor data. The independent training of both autoencoders from normalized data emerges as a prudent design choice, preventing the propagation of reconstruction errors while maintaining modularity. Additionally, the joint fine-tuning strategy proves instrumental in achieving strong generalization with a unified global classifier, eliminating the need for maintaining separate chip-specific models.

The framework demonstrates robust performance across chips with varying fabrication characteristics, achieving consistent classification accuracy while maintaining excellent

generalization from training to test data. The temperature compensation autoencoder successfully mitigates thermal variations, while the fabrication error compensation autoencoder effectively aligns diverse chip responses to a common representation space. The near-perfect detection of no-gas conditions confirms the preservation of discriminative spectral features throughout the compensation pipeline.

Looking into the future, there exists substantial potential for further research. Refining the autoencoder training process by employing unsupervised or semi-supervised learning approaches could reduce the labeled data requirements, particularly for the fabrication compensation stage which currently necessitates matched reference pairs. Experimentation with transfer learning techniques for rapid adaptation to new chips might unveil efficient calibration methods, minimizing the burden of collecting extensive training data for each deployed sensor.

Moreover, extending the framework to accommodate more diverse environmental conditions, including broader temperature ranges and additional confounding factors, could enhance the robustness and applicability of the approach. Investigation into alternative autoencoder architectures, such as Variational Autoencoders or attention-based mechanisms, might provide uncertainty quantification capabilities and capture more complex relationships between environmental factors and spectral distortions.

The development of model compression and optimization strategies would facilitate deployment on resource-constrained edge devices, enabling real-time environmental monitoring in distributed sensor networks. Furthermore, extending the classification framework to support quantitative concentration estimation and temporal trend analysis could provide more comprehensive environmental assessment capabilities beyond discrete class identification.

Lastly, examining the application of this methodology to other photonic sensing modalities experiencing similar fabrication and environmental sensitivities could yield insights into the framework's versatility and broader applicability across diverse spectroscopic analysis domains.

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